

Matlacha Pass Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

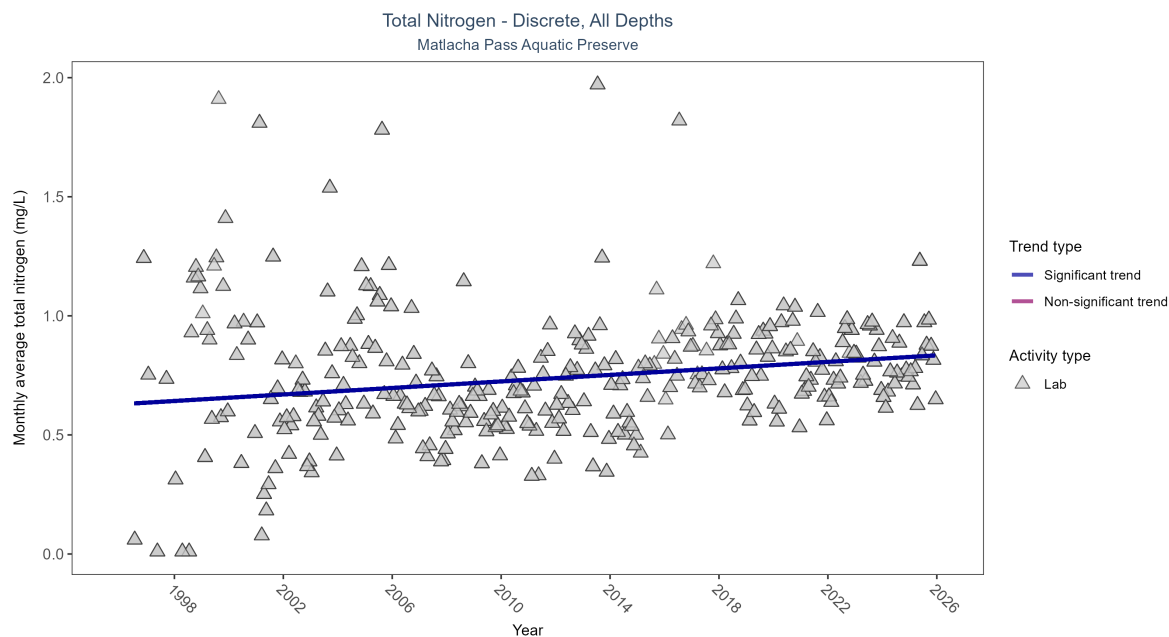


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	1589	30	1996 - 2025	0.724	0.17111	0.62869	0.00685	0

Monthly average total nitrogen increased by 0.01 mg/L per year.

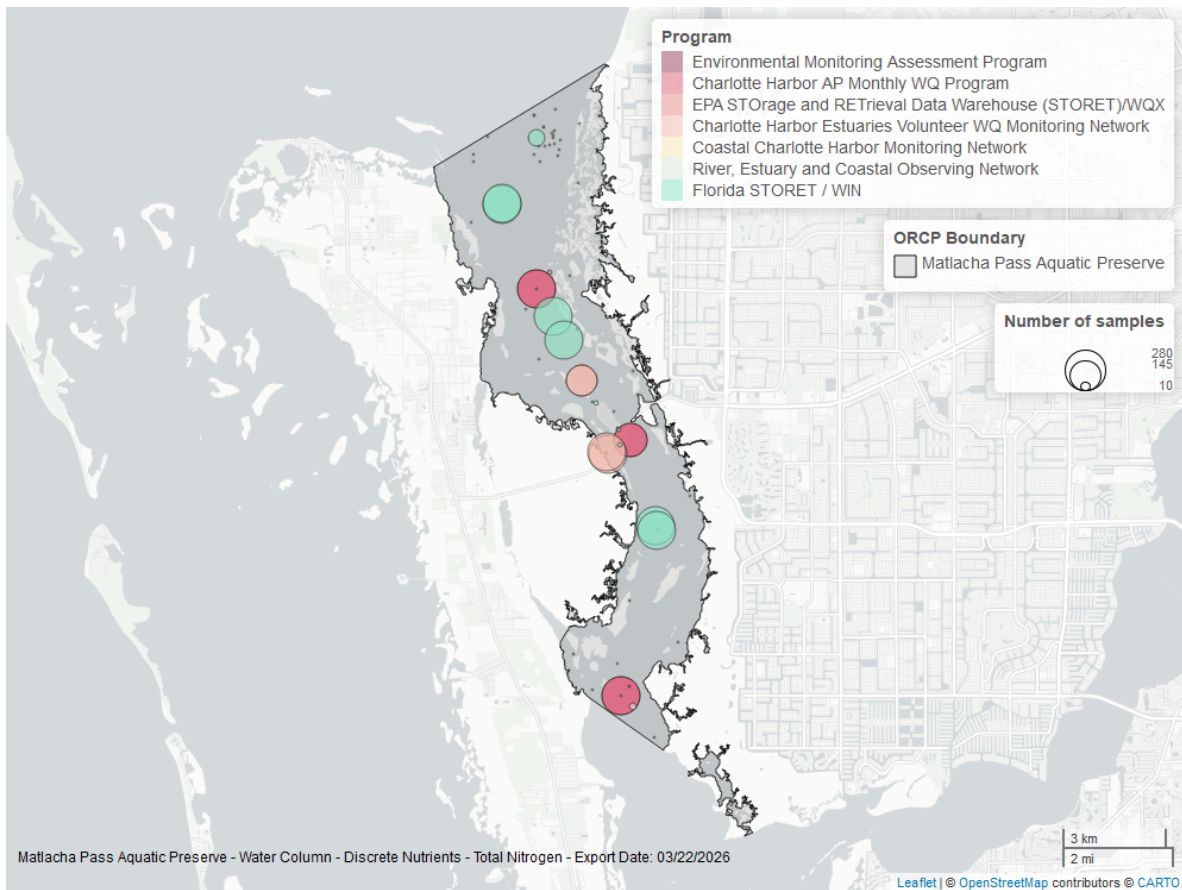


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

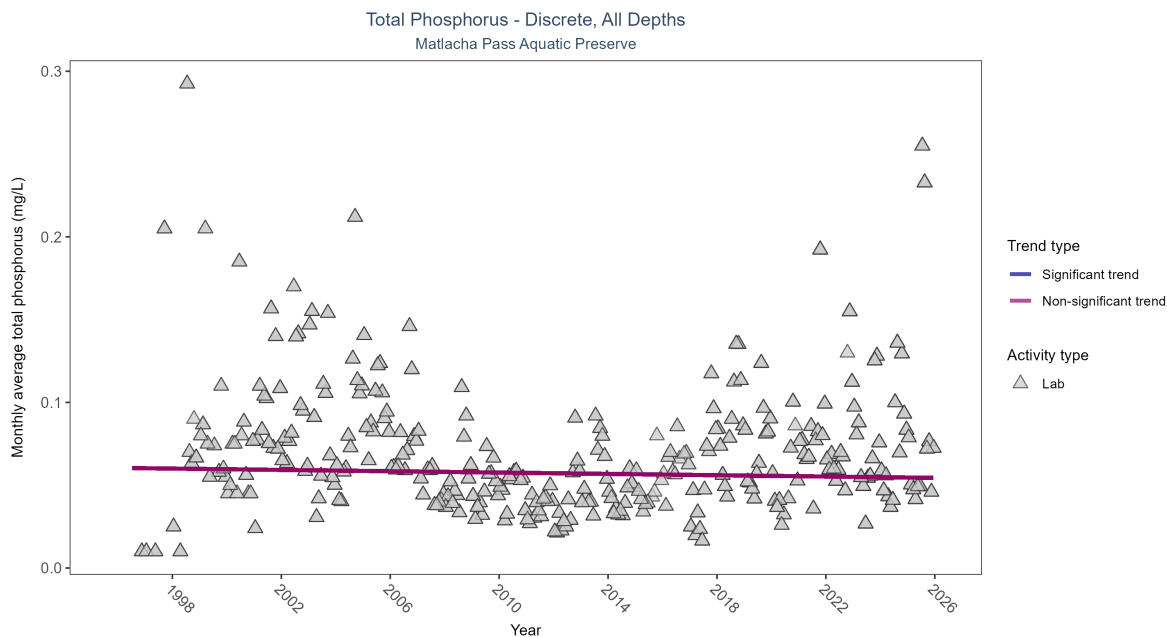


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	2095	30	1996 - 2025	0.06	-0.03825	0.06044	-0.0002	0.378

Total phosphorus showed no detectable trend between 1996 and 2025.

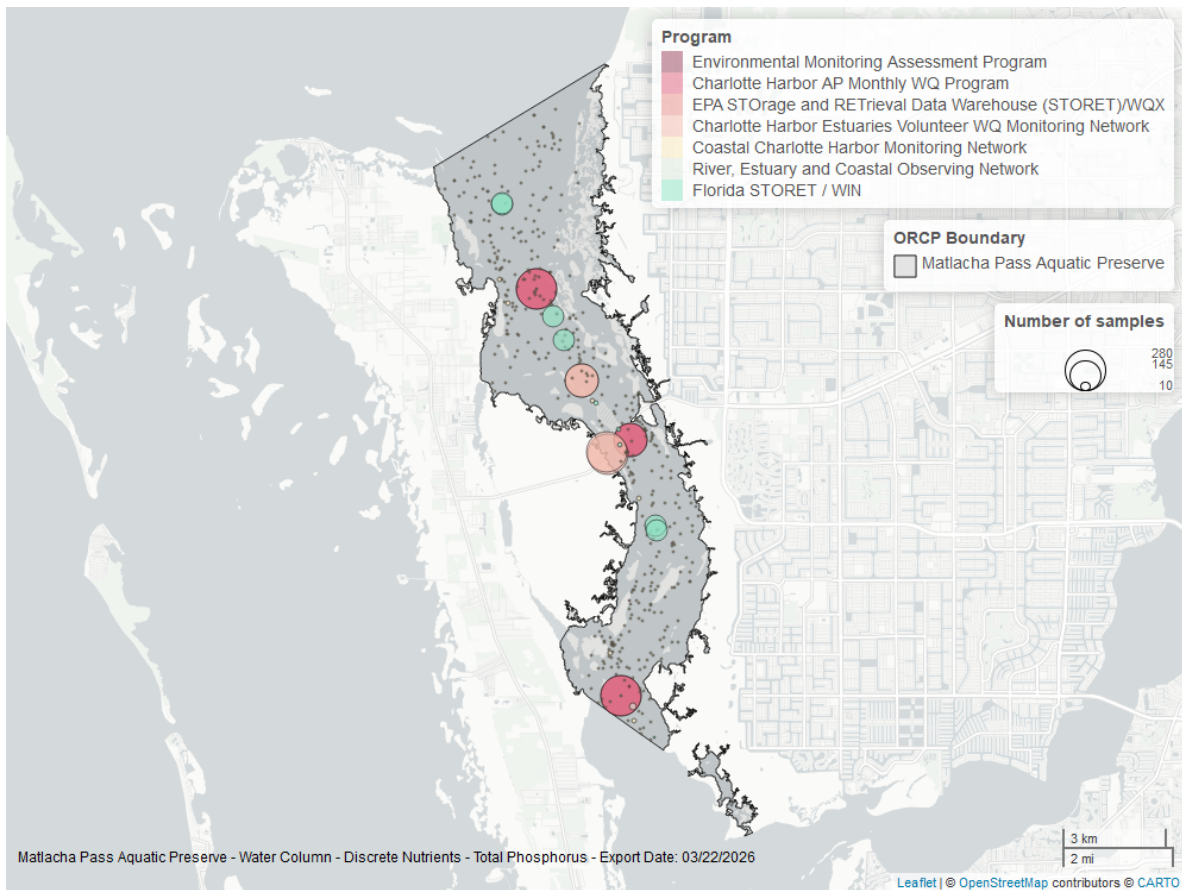


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

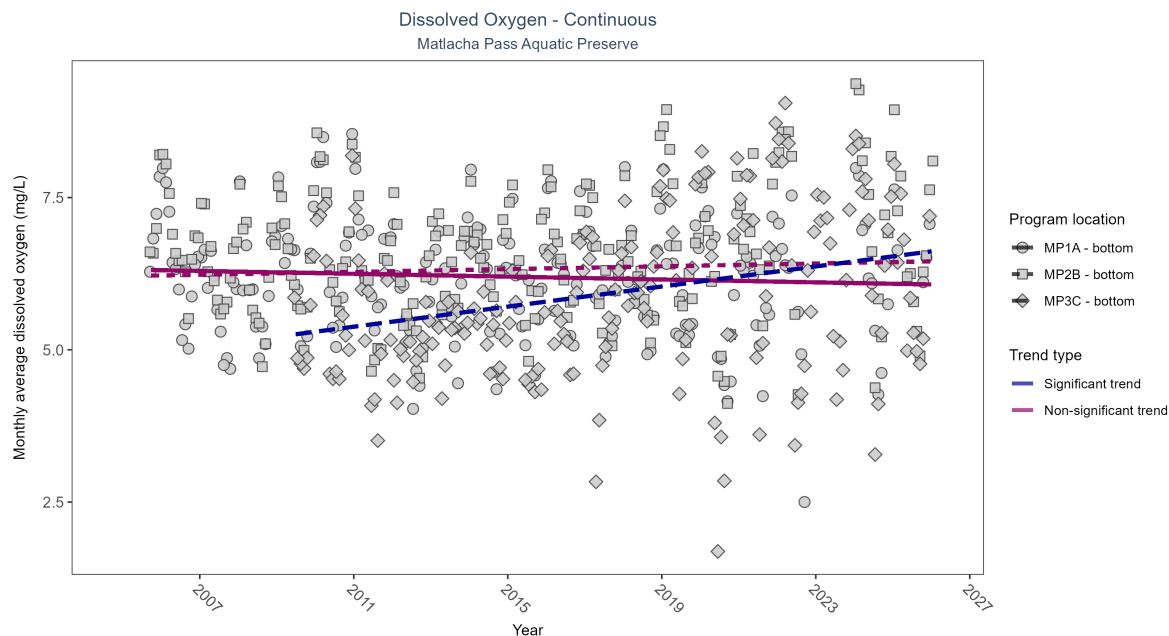


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	No significant trend	564550	20	2005 - 2025	6.3	-0.07	6.32	-0.01	0.1745
MP2B	No significant trend	593033	21	2005 - 2026	6.7	0.08	6.21	0.01	0.1021
MP3C	Significantly increasing trend	510586	17	2009 - 2025	5.9	0.28	5.21	0.08	0

At one program location, monthly average dissolved oxygen increased by 0.08 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at two locations.

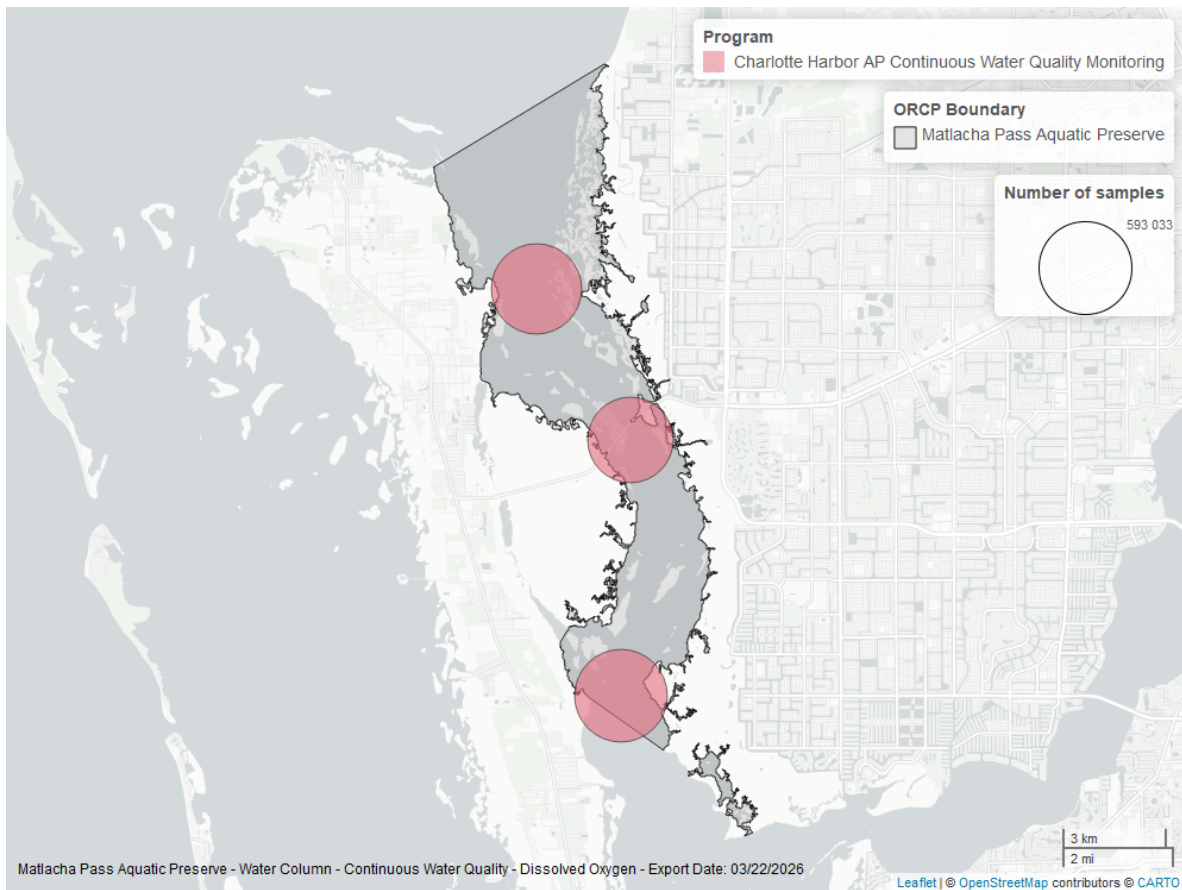


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

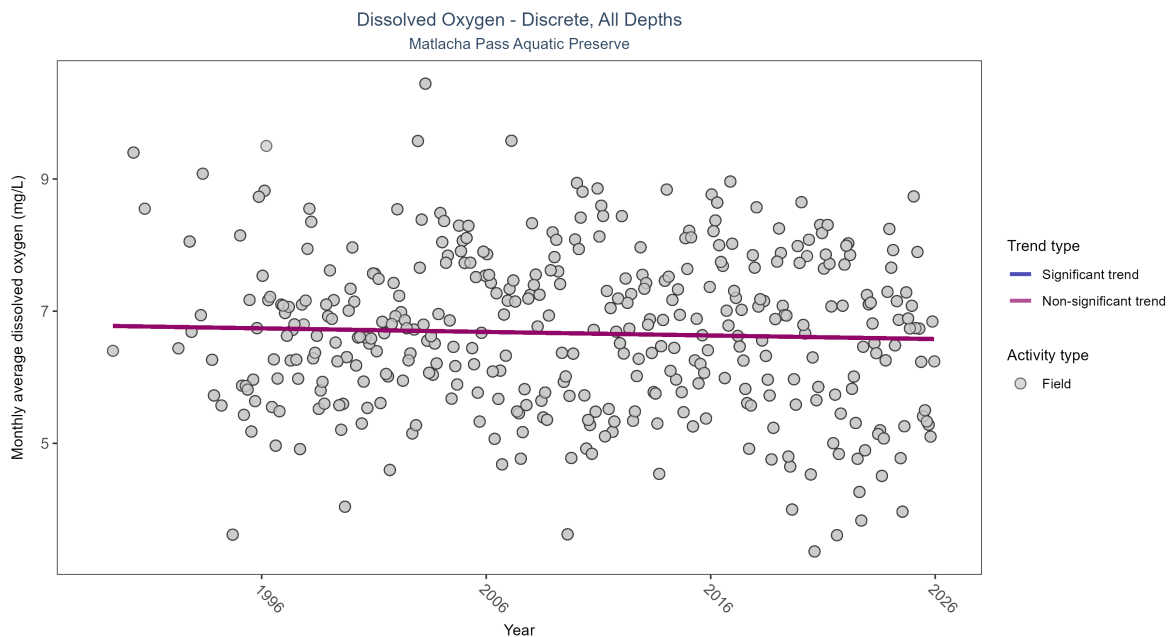


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	10959	36	1989 - 2025	6.6	-0.04416	6.77759	-0.0054	0.2228

Dissolved oxygen showed no detectable trend between 1989 and 2025.

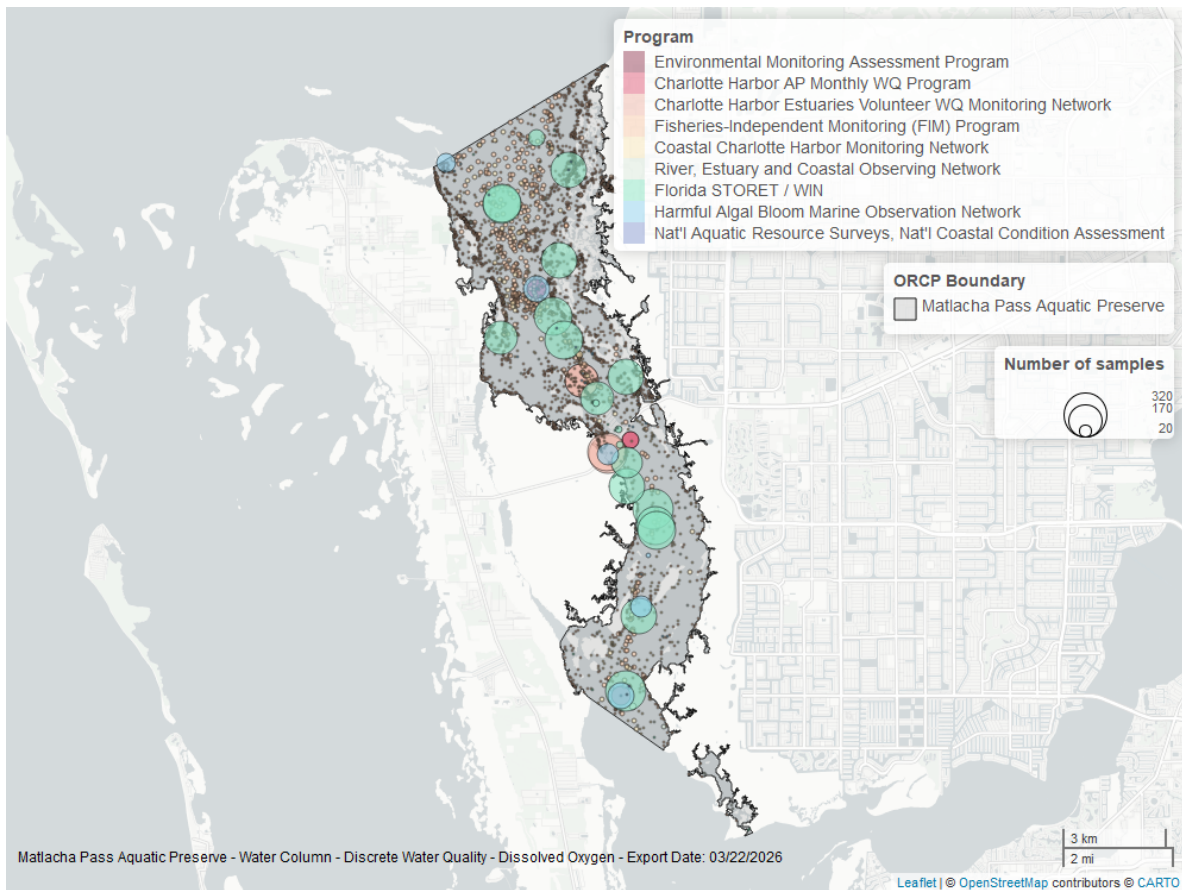


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

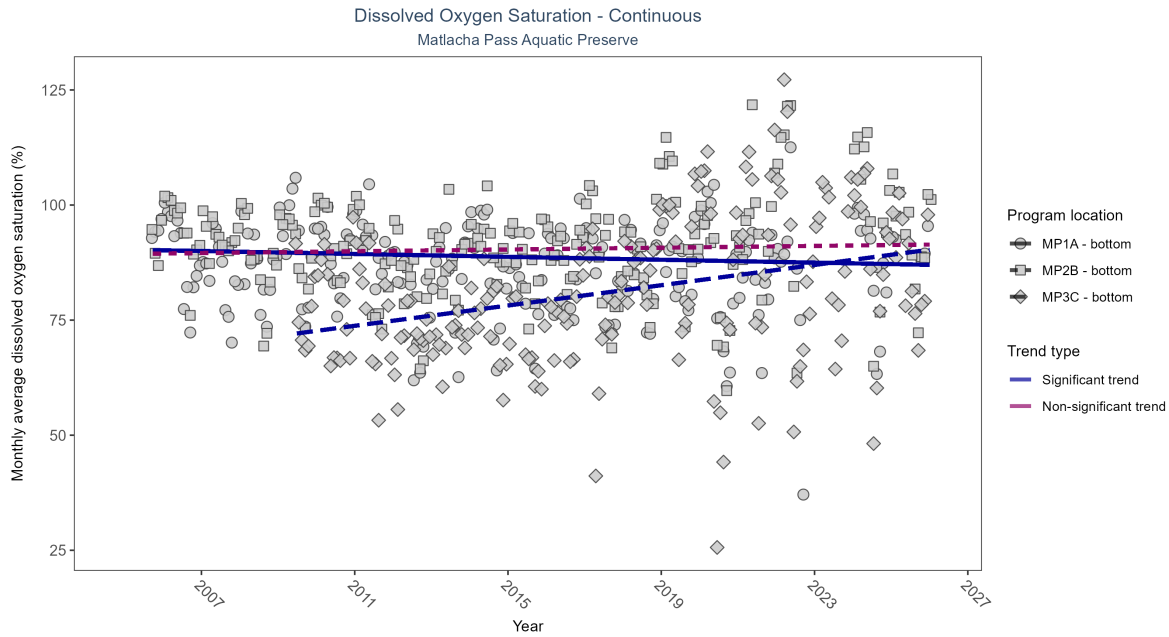


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Significantly decreasing trend	561908	20	2005 - 2025	88.1	-0.11	90.32	-0.16	0.03
MP2B	No significant trend	593334	21	2005 - 2026	91.5	0.05	89.37	0.10	0.3266
MP3C	Significantly increasing trend	514714	17	2009 - 2025	81.8	0.28	71.56	1.10	0

At one program location, monthly average dissolved oxygen saturation increased by 1.10% per year. At one program location, monthly average dissolved oxygen saturation decreased by 0.16% per year. No detectable change in monthly average dissolved oxygen saturation was observed at one location.

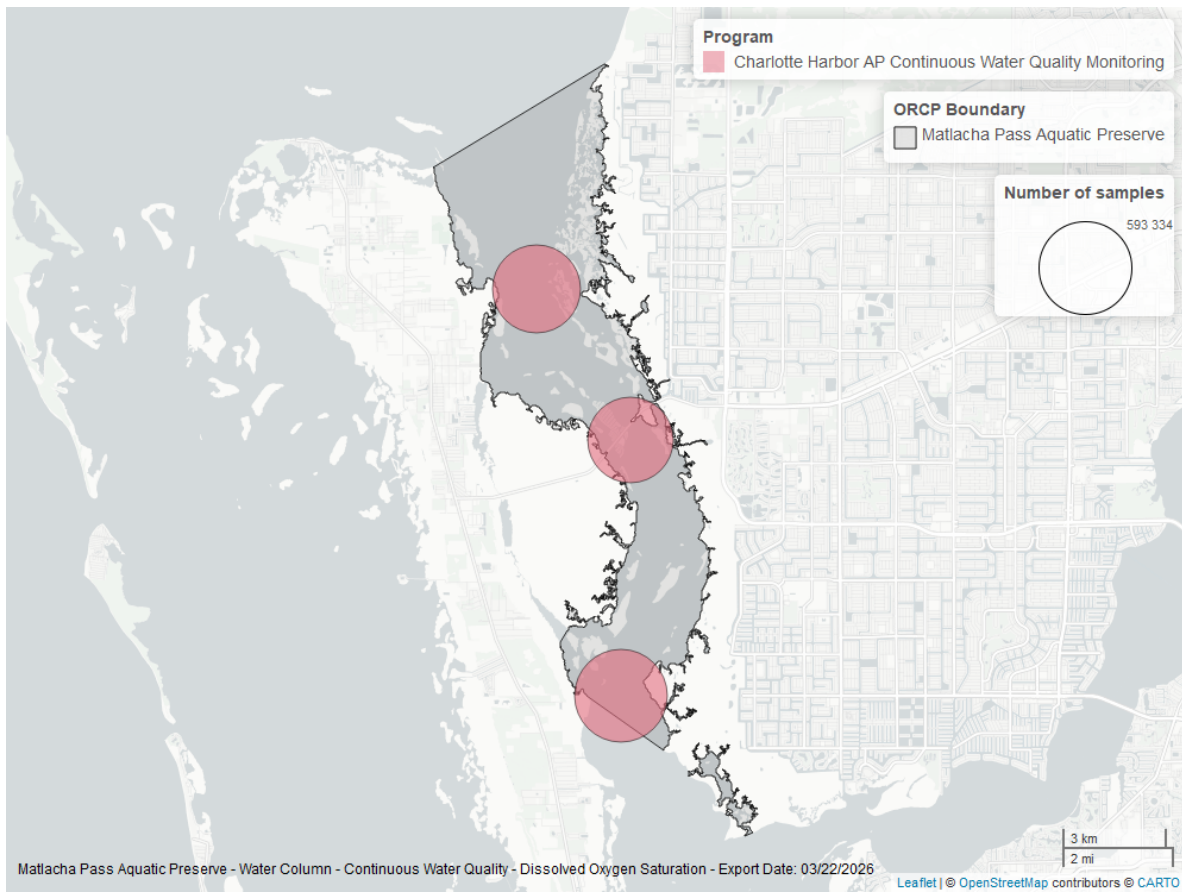


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

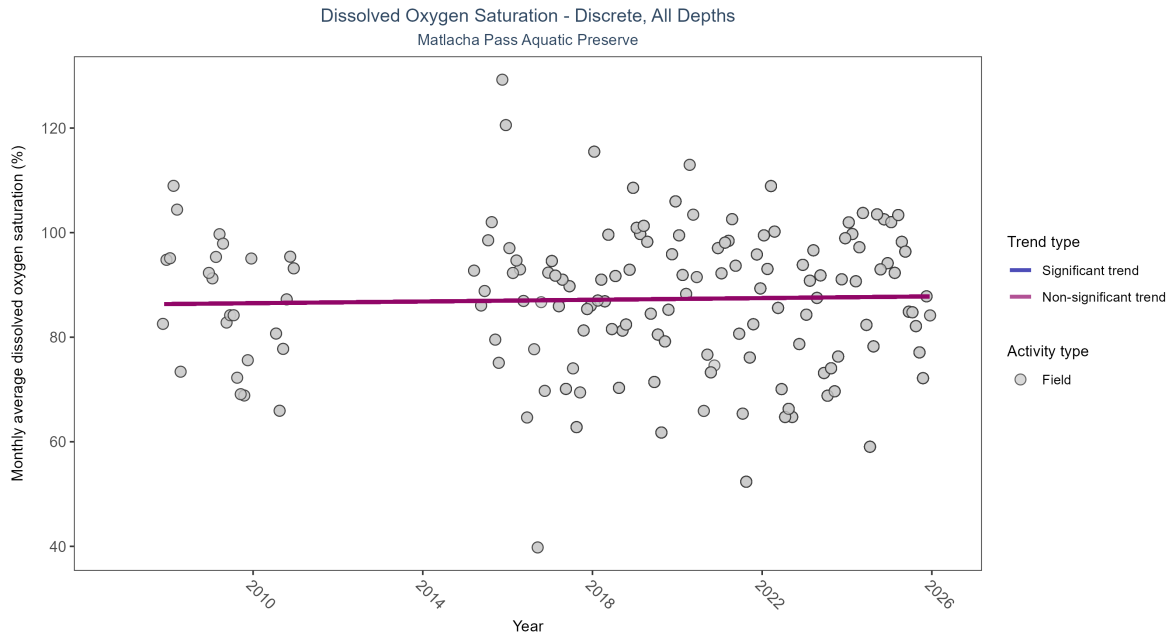


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	1464	15	2007 - 2025	89.80714	0.04683	86.25973	0.08125	0.4532

Dissolved oxygen saturation showed no detectable trend between 2007 and 2025.

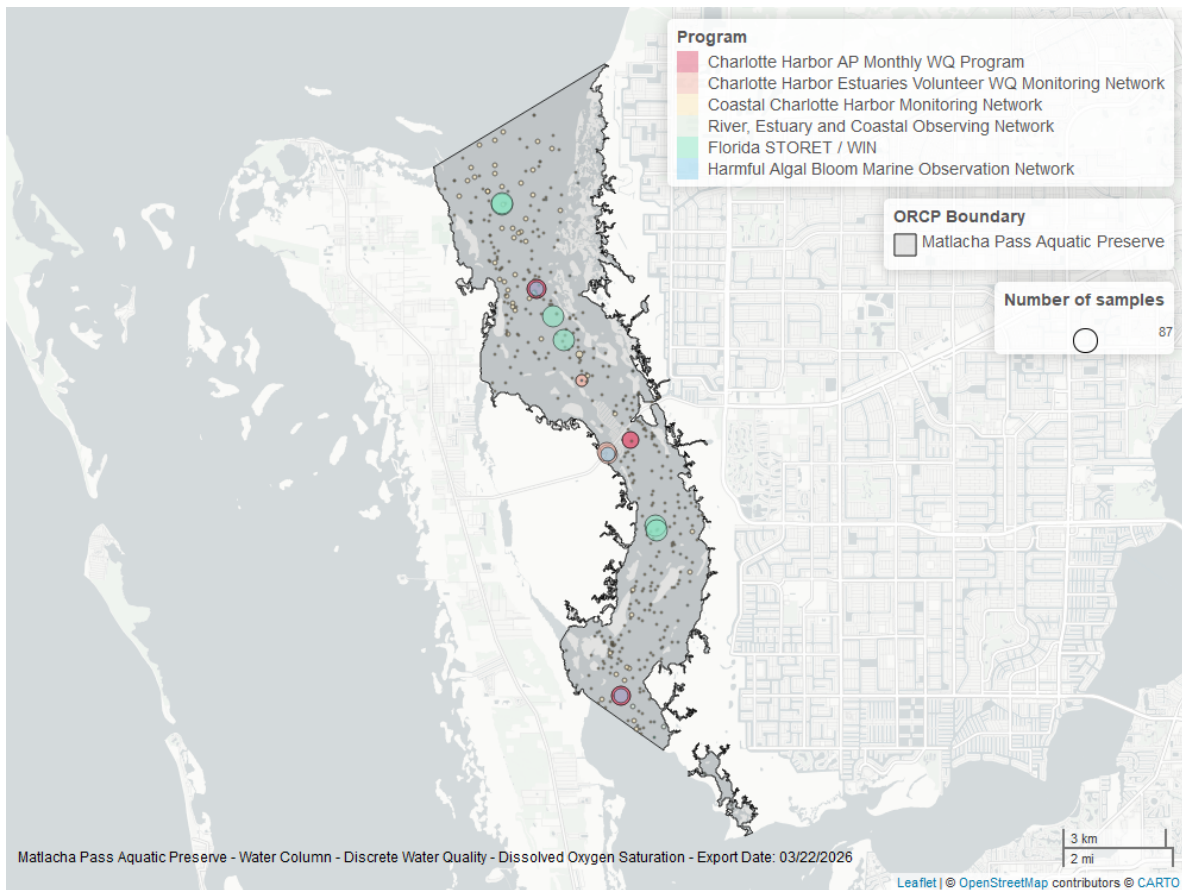


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

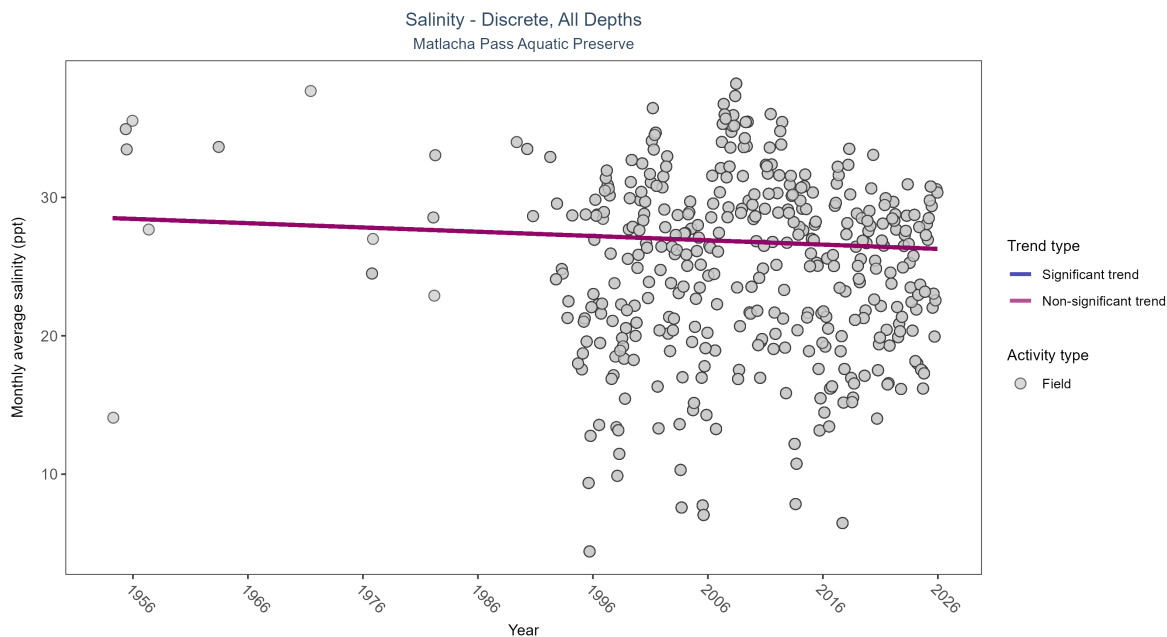


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No significant trend	11207	43	1954 - 2025	25.9	-0.04336	28.50998	-0.03095	0.2072

Salinity showed no detectable trend between 1954 and 2025.

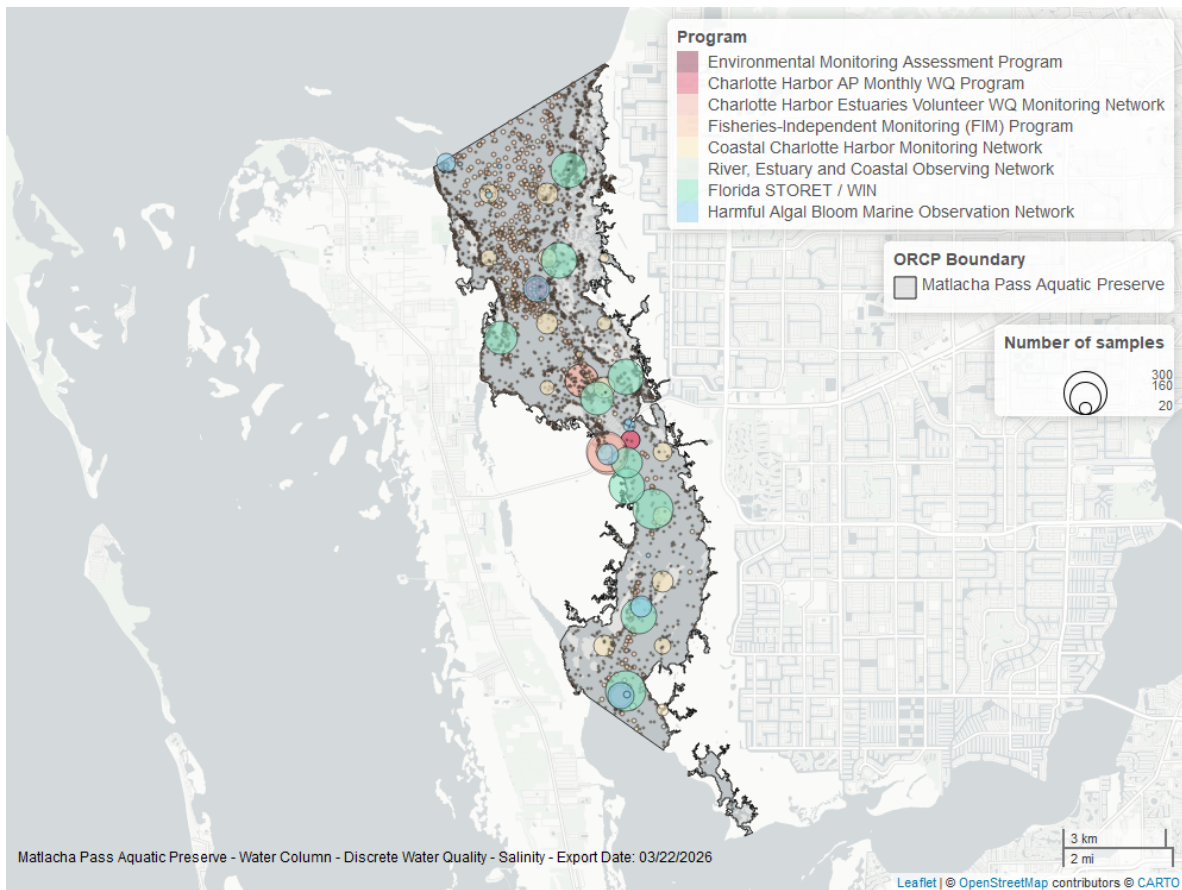


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

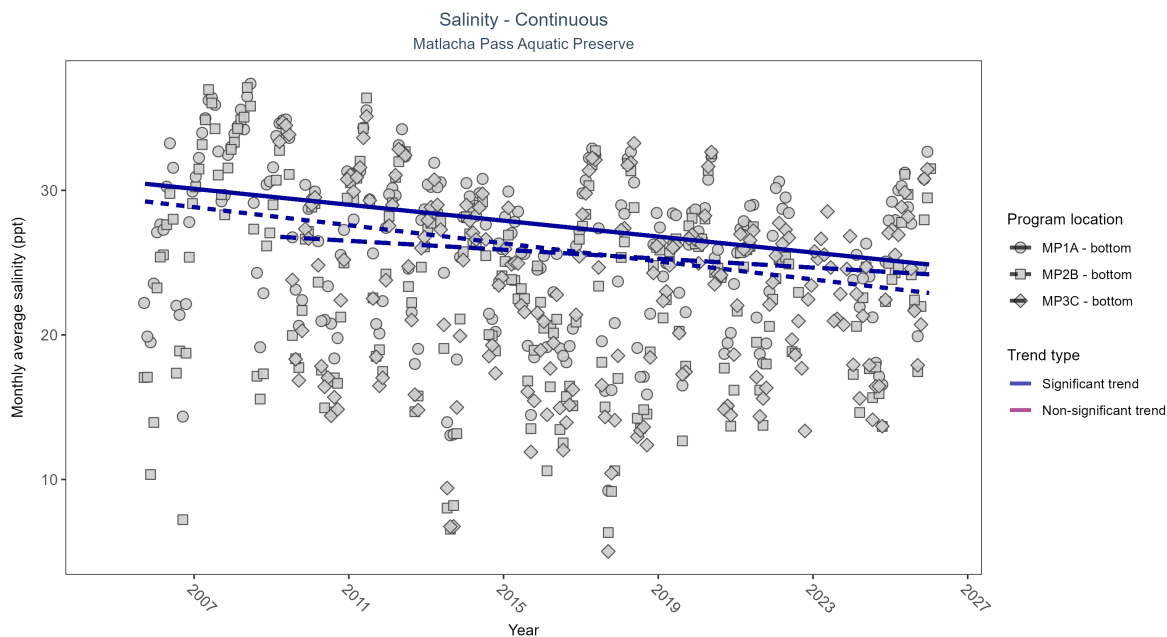


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Significantly decreasing trend	604162	20	2005 - 2025	27.1	-0.29	30.66	-0.28	0
MP2B	Significantly decreasing trend	625170	21	2005 - 2026	24.4	-0.27	29.46	-0.31	0
MP3C	Significantly decreasing trend	549532	17	2009 - 2025	23.7	-0.12	26.82	-0.15	0.0213

At three program locations, monthly average salinity decreased between 0.15 and 0.31 ppt per year.

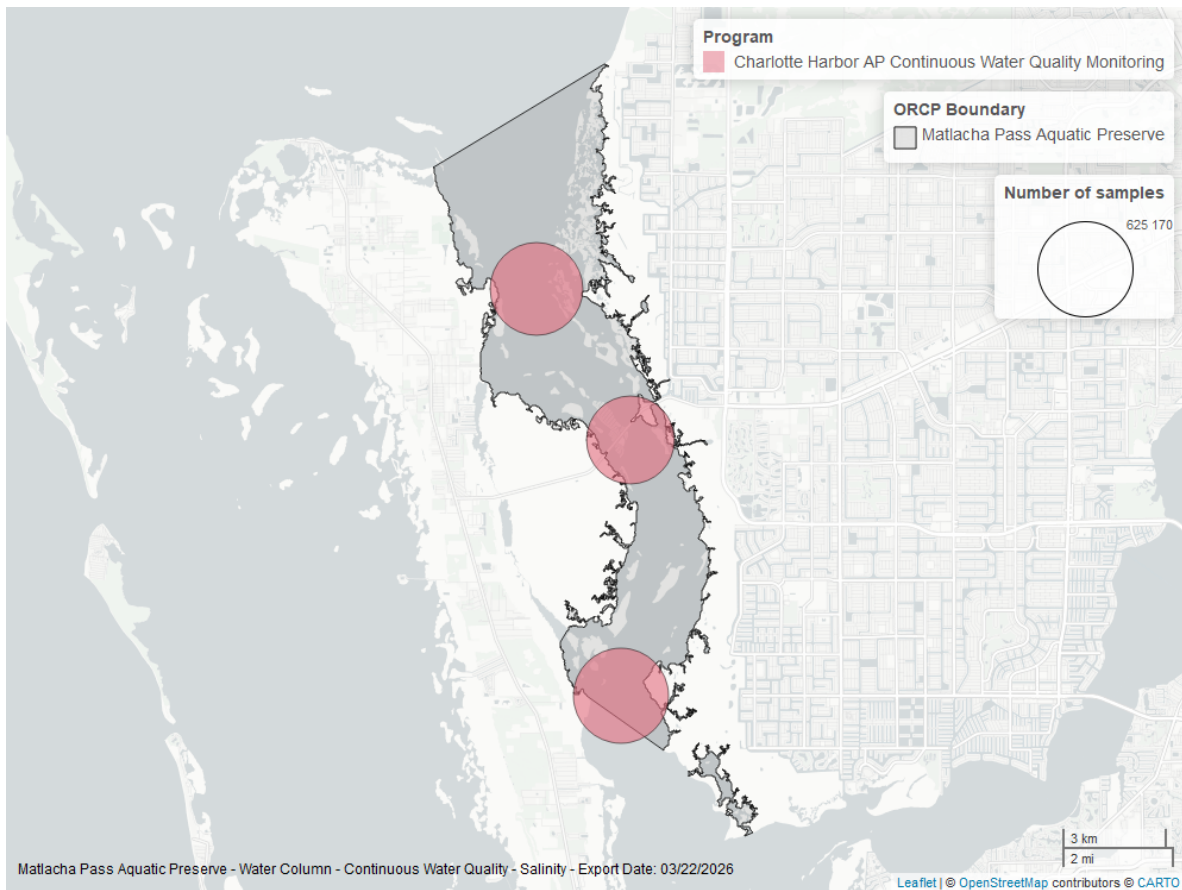


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

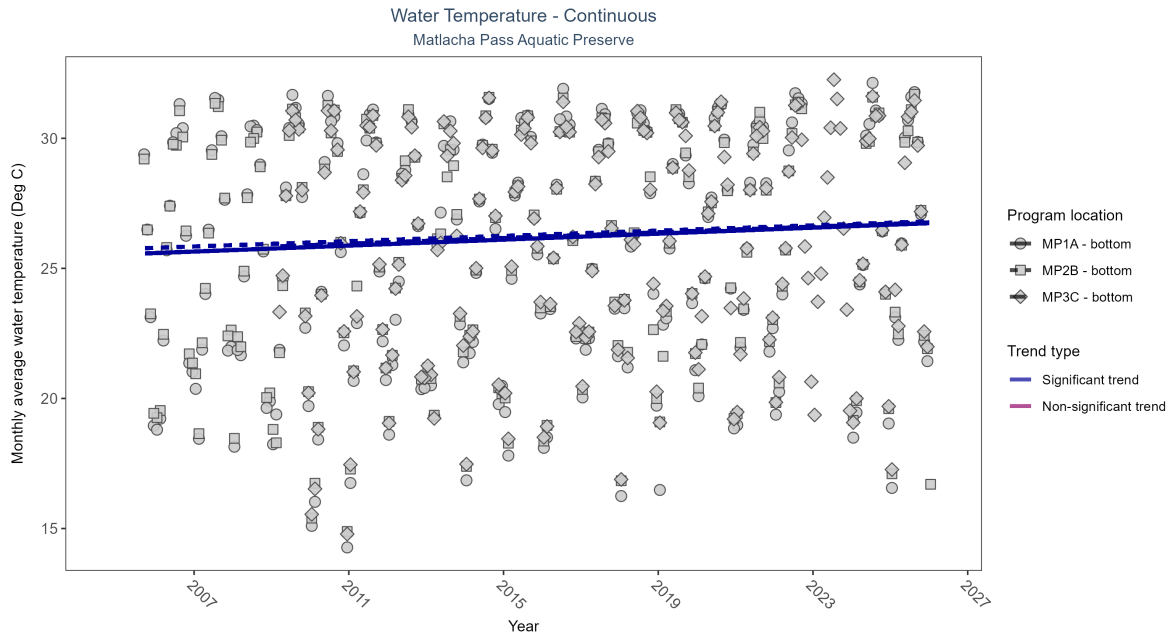


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Significantly increasing trend	643697	20	2005 - 2025	26.4	0.25	25.53	0.06	0
MP2B	Significantly increasing trend	639164	21	2005 - 2026	26.2	0.21	25.74	0.05	0
MP3C	Significantly increasing trend	557705	17	2009 - 2025	26.6	0.20	25.84	0.05	0.0001

At three program locations, monthly average water temperature increased between 0.05 and 0.06°C per year.

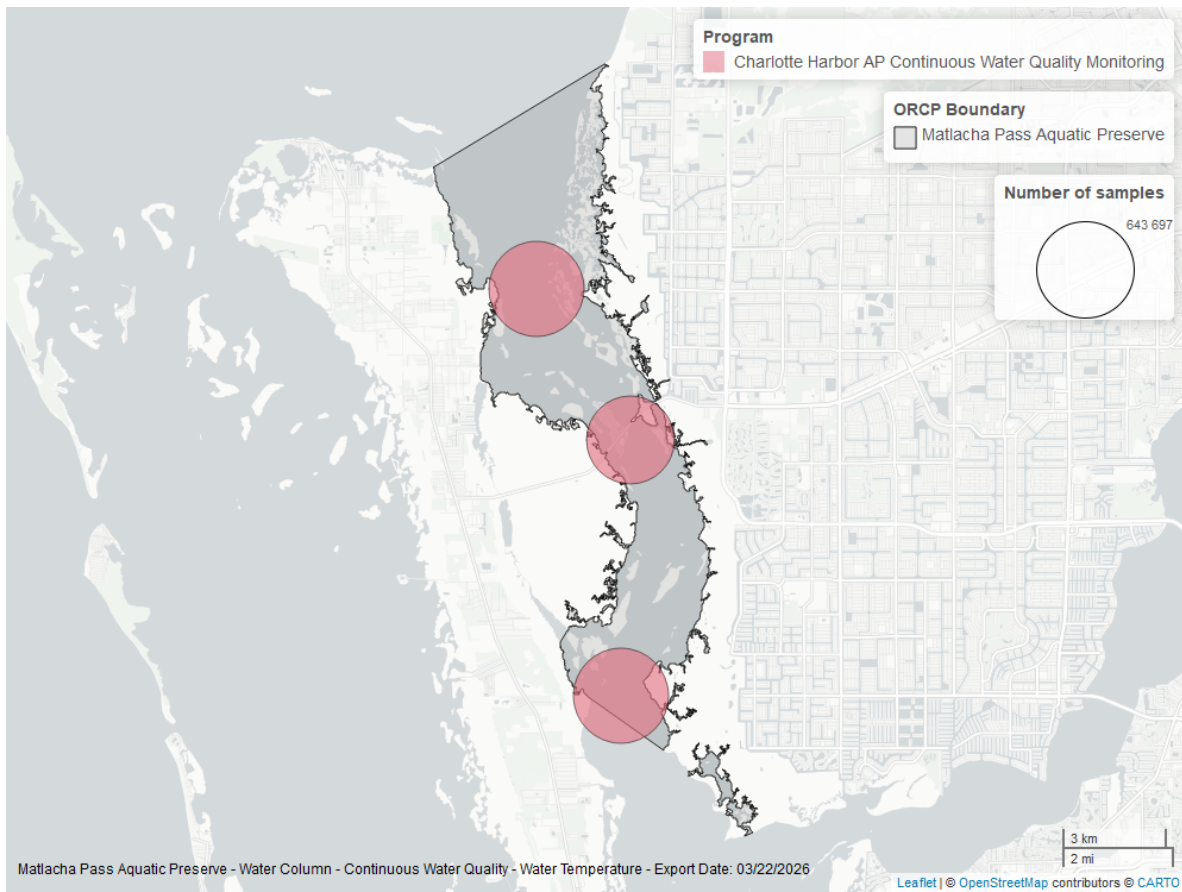


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

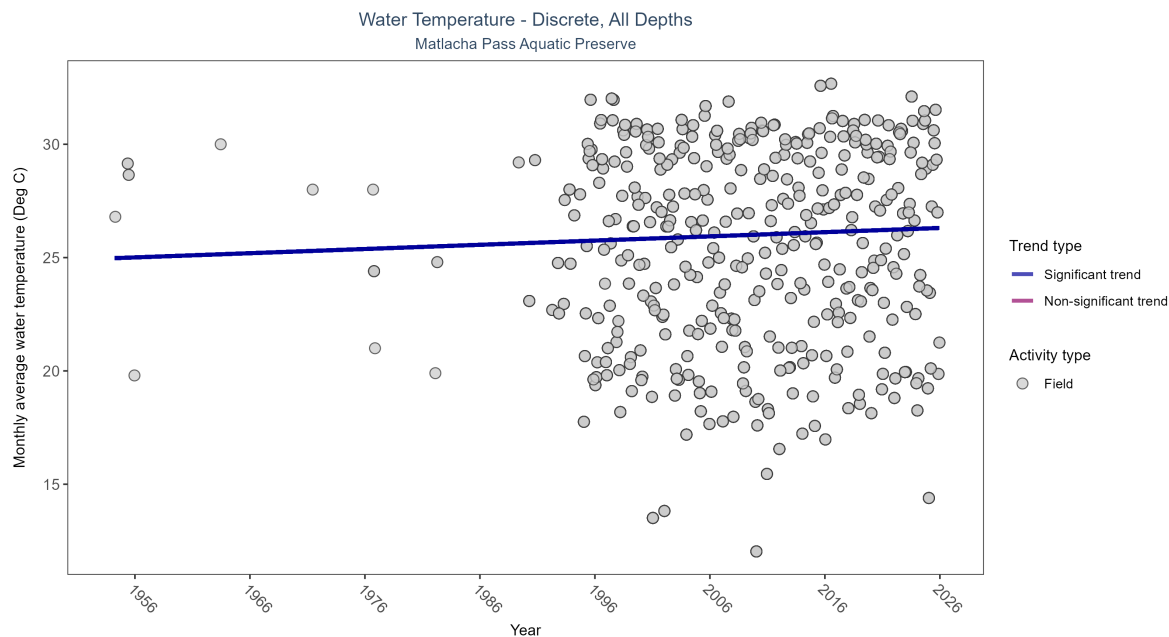


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	11178	42	1954 - 2025	26.3	0.09361	24.97043	0.01857	0.0067

Monthly average water temperature increased by 0.02°C per year.

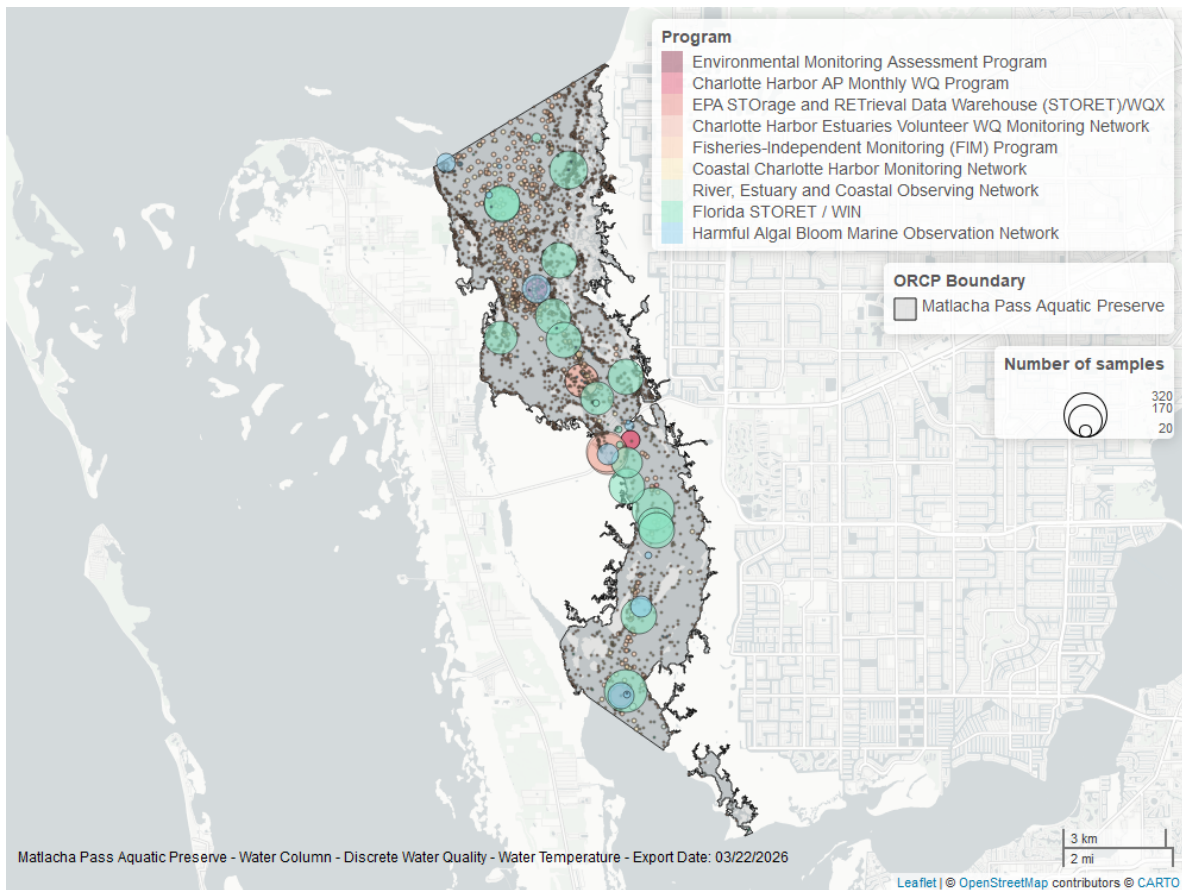


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

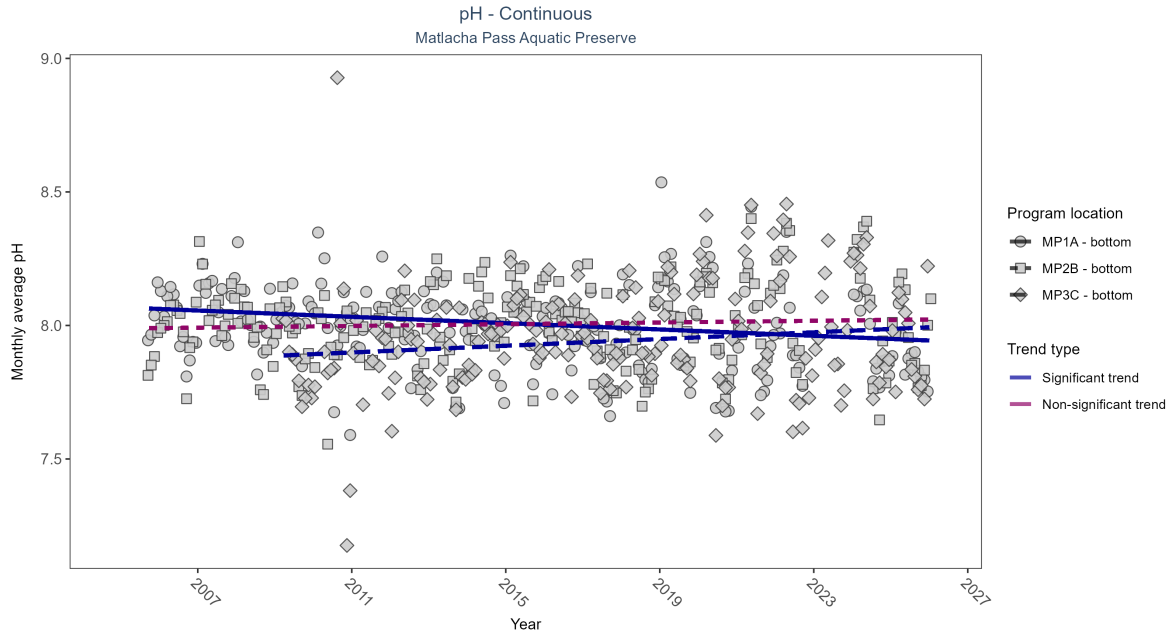


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Significantly decreasing trend	556253	20	2005 - 2025	8.0	-0.19	8.07	-0.01	0.0001
MP2B	No significant trend	573184	21	2005 - 2026	8.0	0.04	7.99	0.00	0.3937
MP3C	Significantly increasing trend	495721	17	2009 - 2025	7.9	0.13	7.89	0.01	0.0186

At one program location, monthly average pH increased by 0.01 pH units per year. At one program location, monthly average pH decreased by 0.01 pH units per year. No detectable change in monthly average pH was observed at one location.

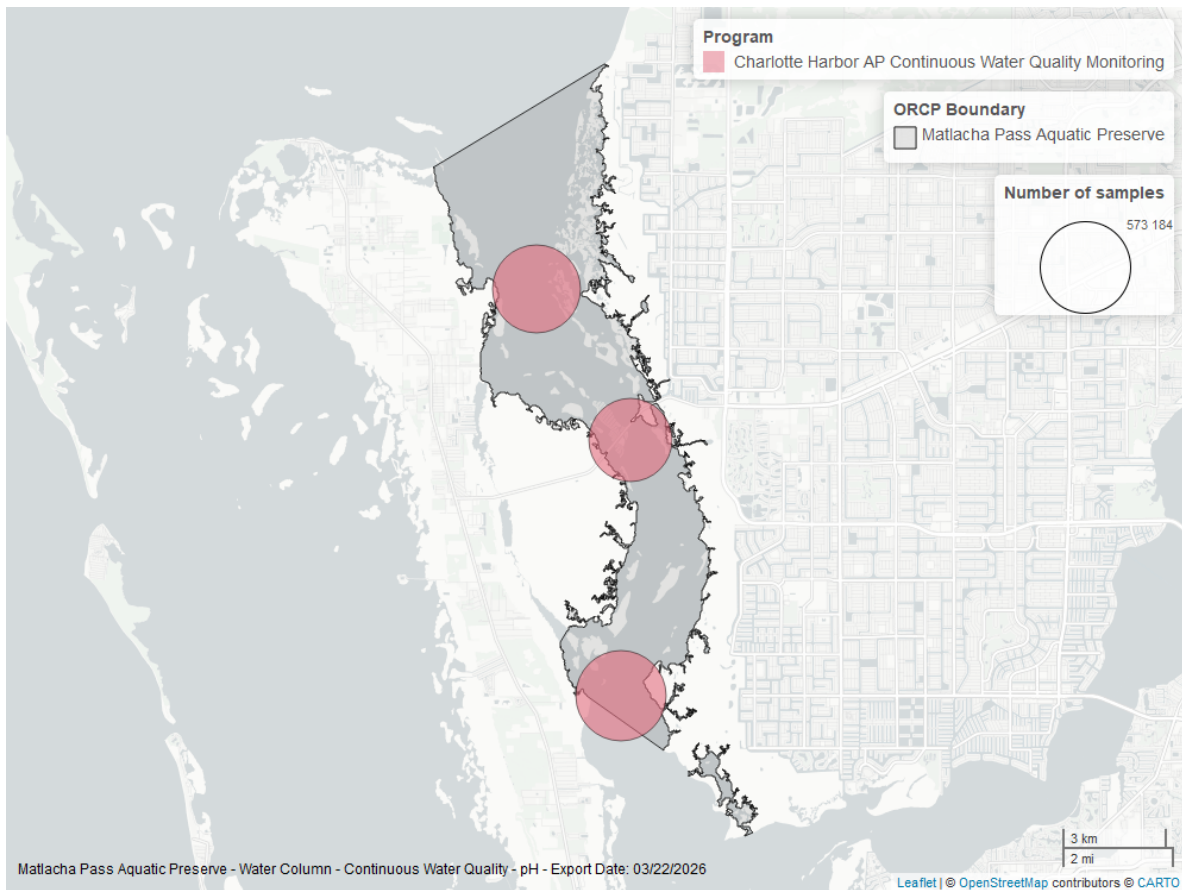


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

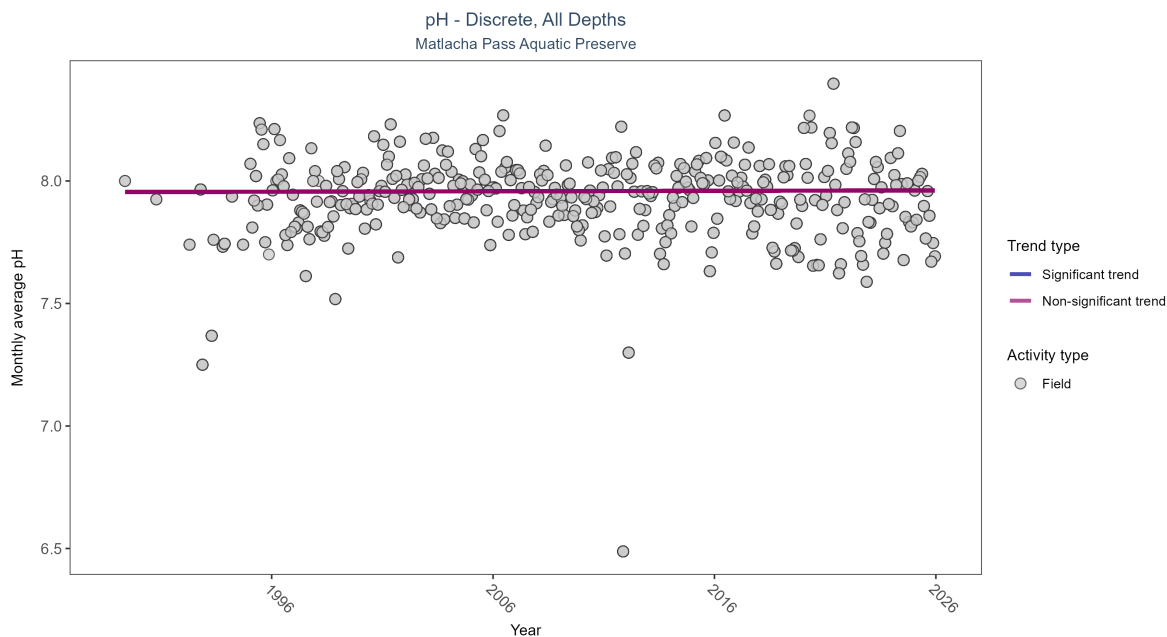


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	10299	36	1989 - 2025	7.92	0.00437	7.95475	0.00018	0.8322

pH showed no detectable trend between 1989 and 2025.

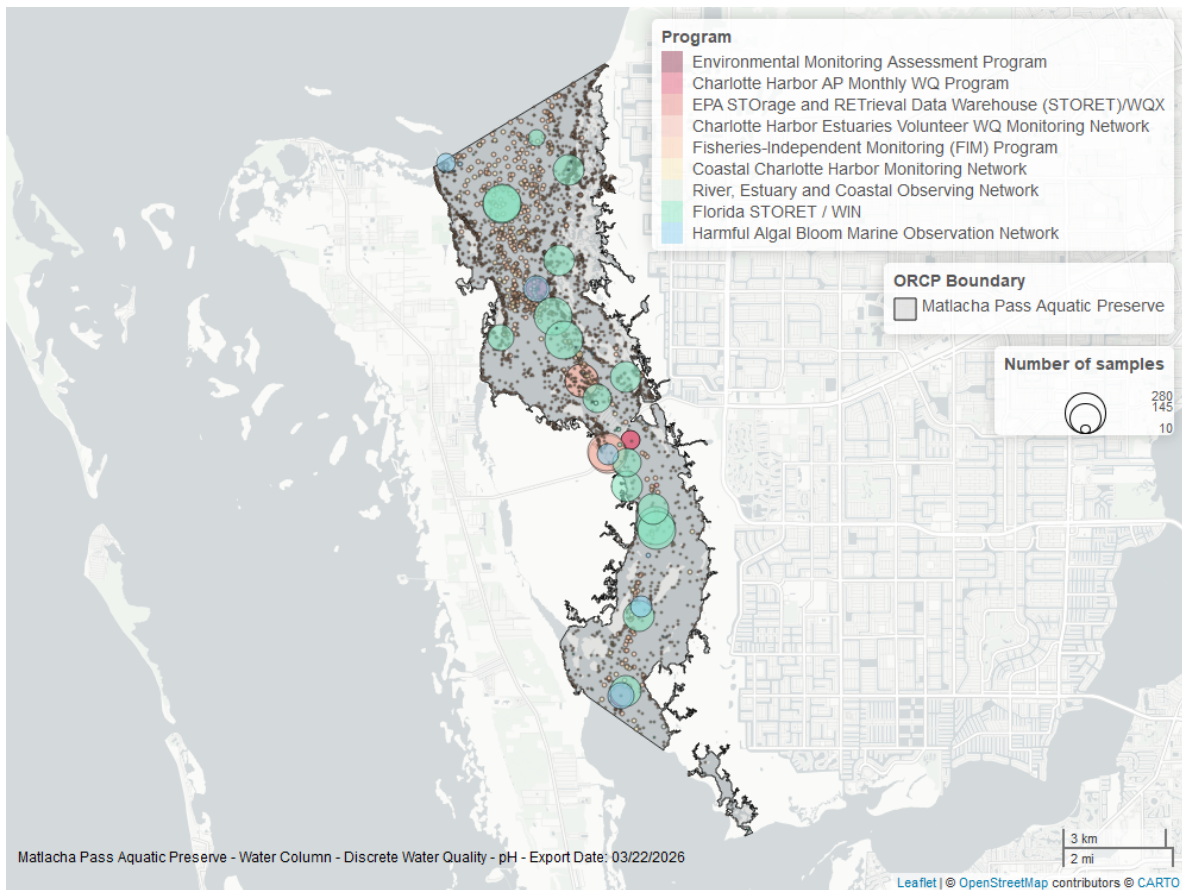


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

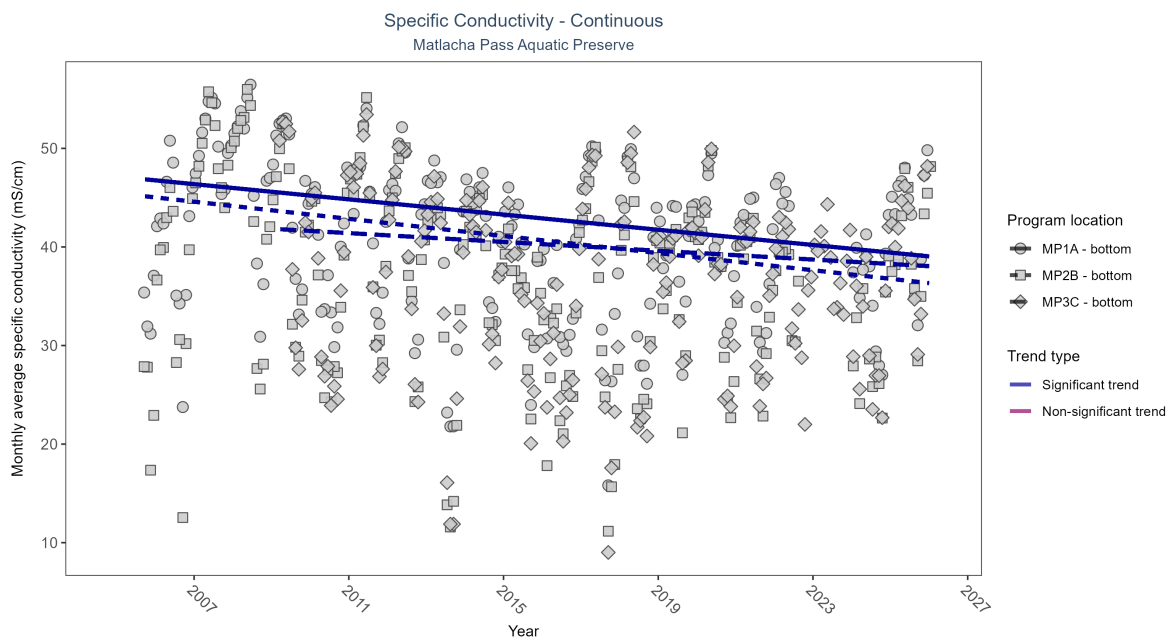


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Significantly decreasing trend	601486	20	2005 - 2025	42.19	-0.29	47.14	-0.39	0
MP2B	Significantly decreasing trend	625170	21	2005 - 2026	38.39	-0.27	45.46	-0.43	0
MP3C	Significantly decreasing trend	548966	17	2009 - 2025	37.50	-0.12	41.85	-0.22	0.0291

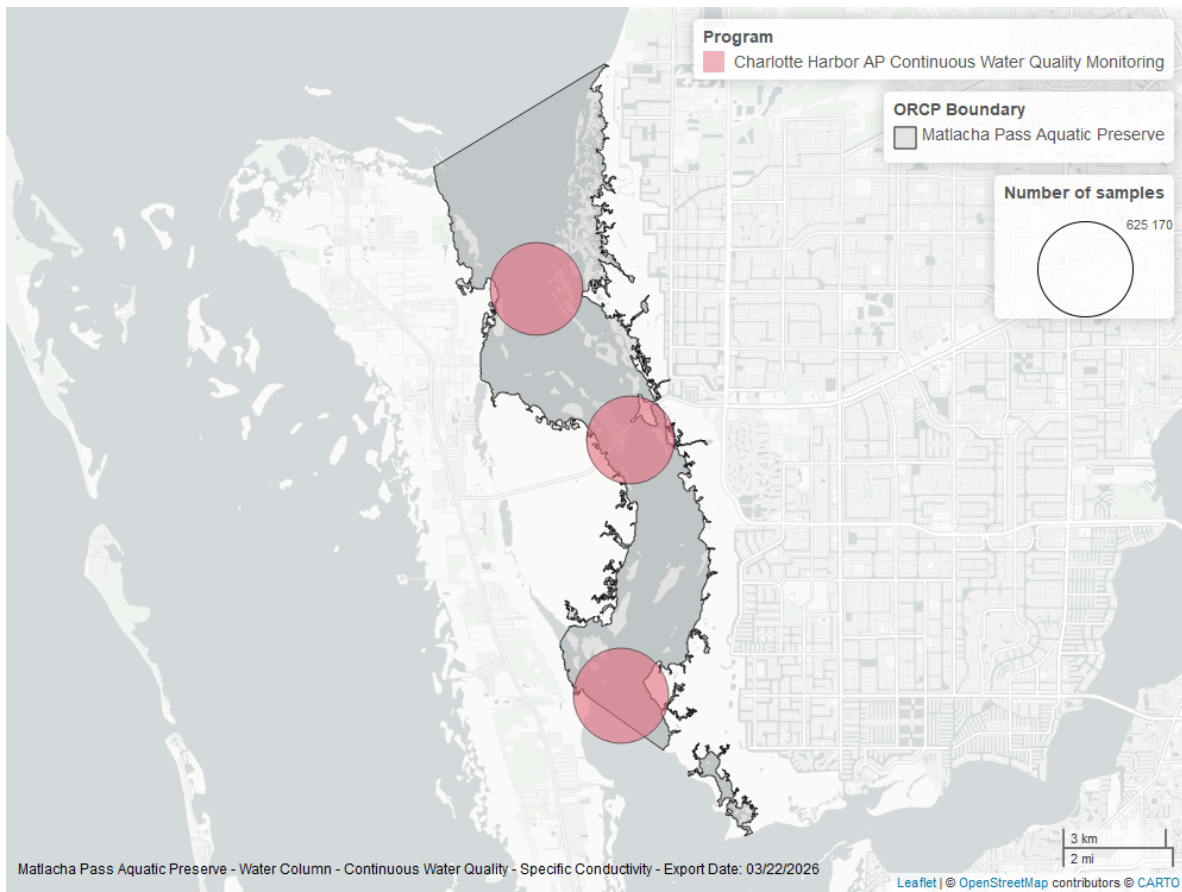


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

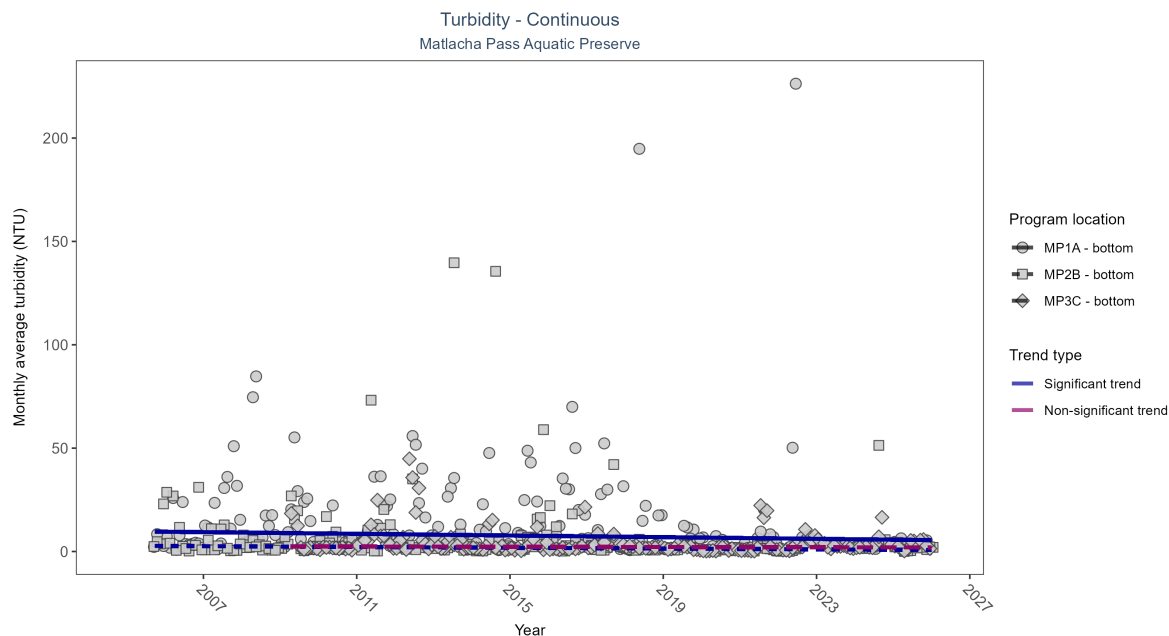


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
MP1A	Significantly decreasing trend	478541	20	2005 - 2025	2	-0.19	9.76	-0.20	0.0001
MP2B	Significantly decreasing trend	537901	21	2005 - 2026	1	-0.18	2.77	-0.10	0.0001
MP3C	No significant trend	478258	17	2009 - 2025	2	-0.06	2.48	-0.03	0.2486

At two program locations, monthly average turbidity decreased by 0.1 NTU per year at one site and by 0.2 NTU per year at the other. No detectable change in monthly average turbidity was observed at one location.

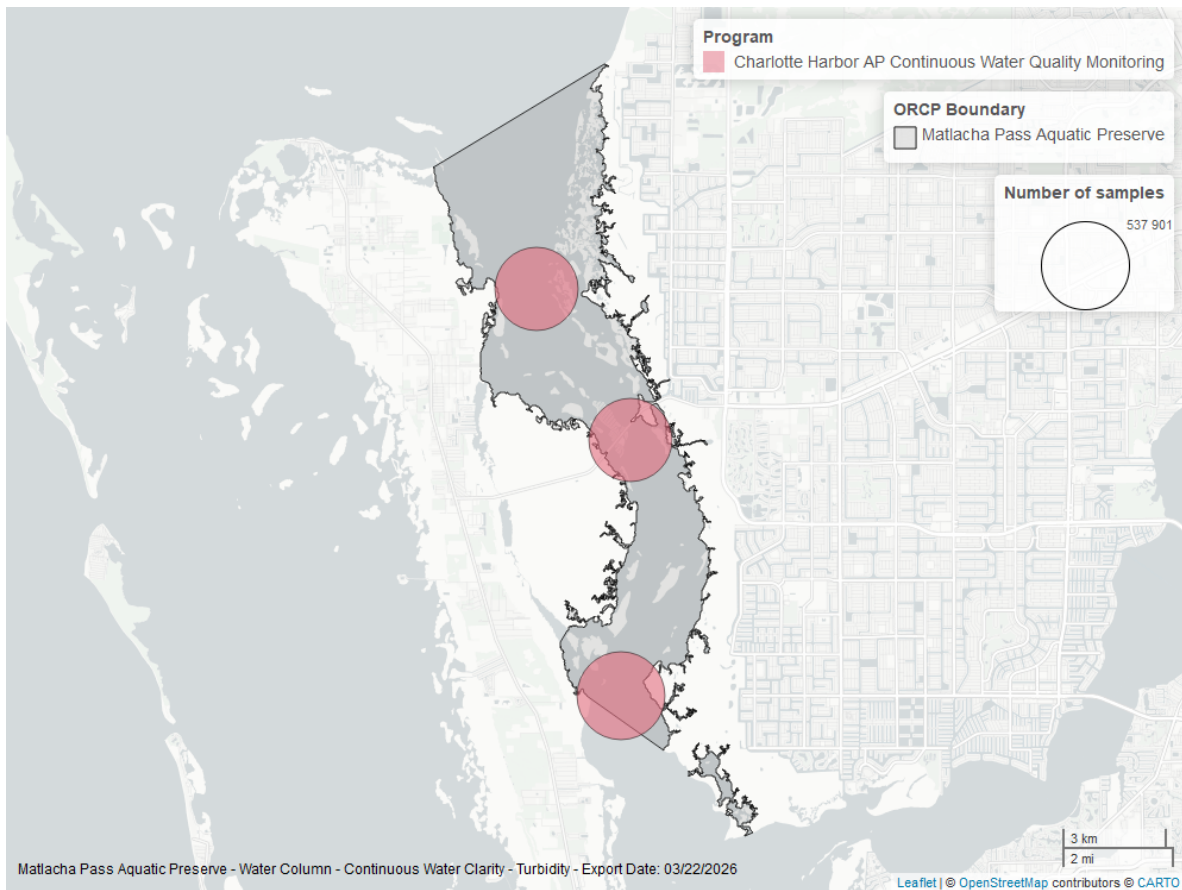


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

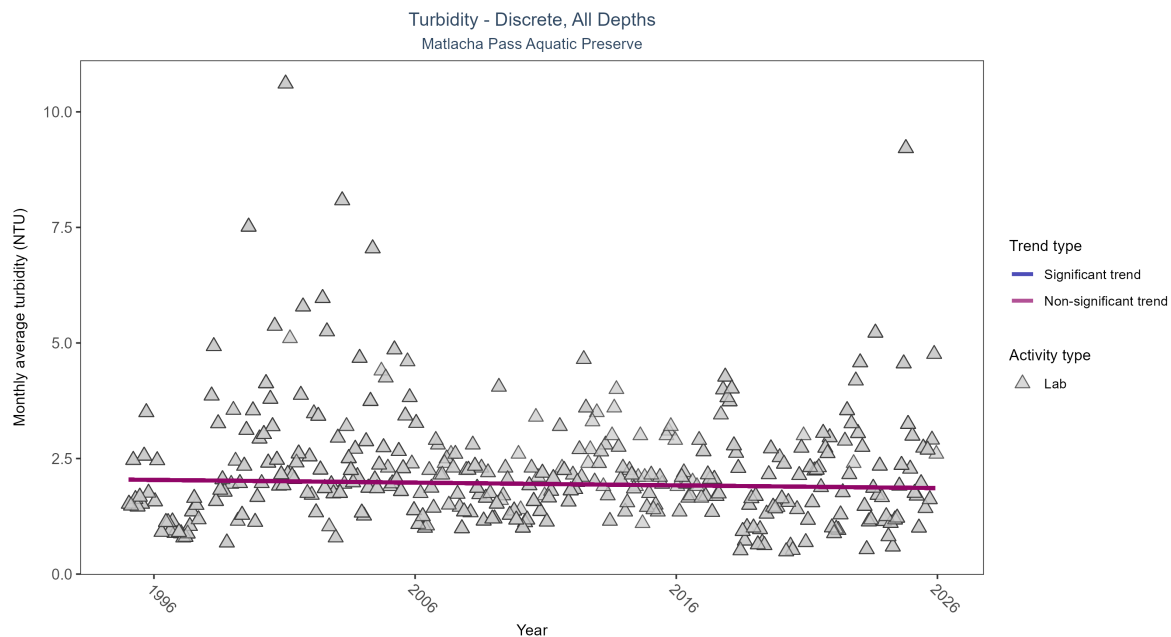


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	2811	31	1995 - 2025	1.8	-0.02878	2.04333	-0.0059	0.3676

Turbidity showed no detectable trend between 1995 and 2025.

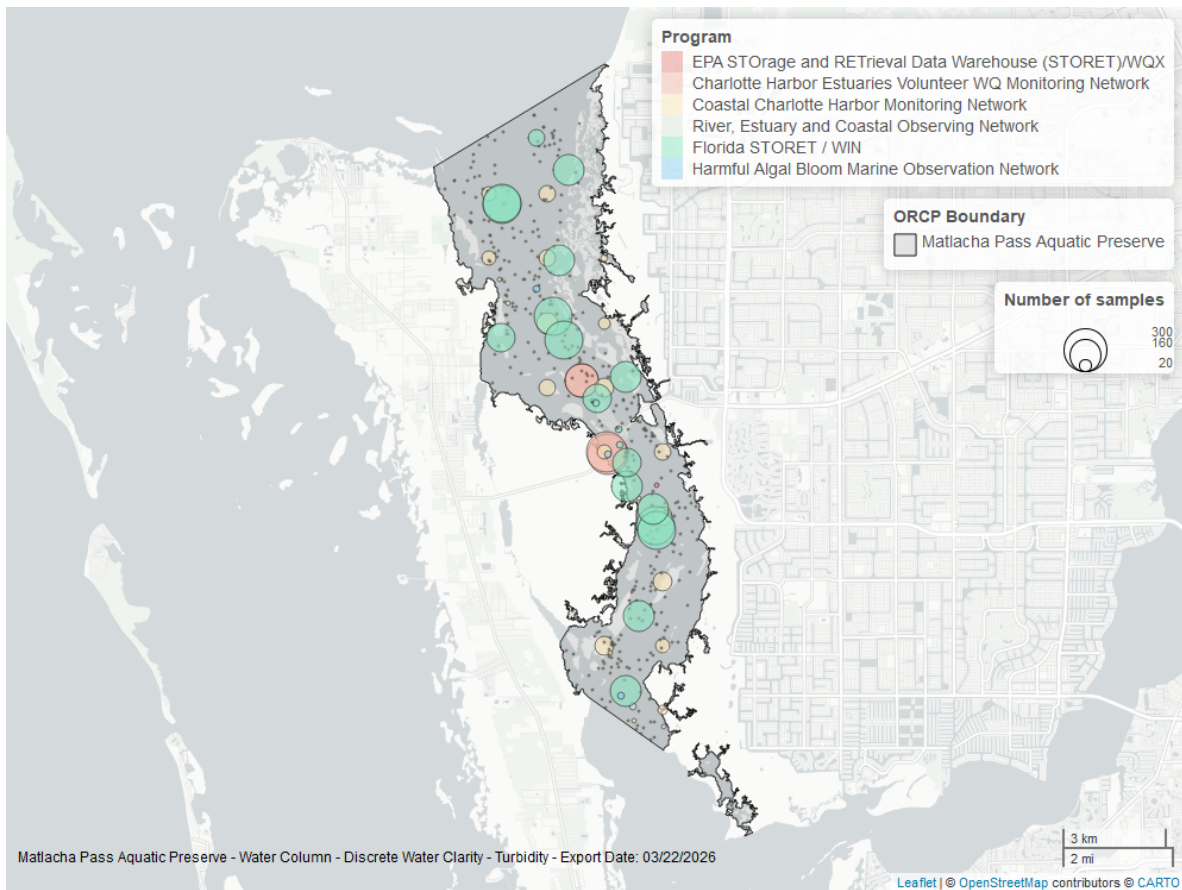


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

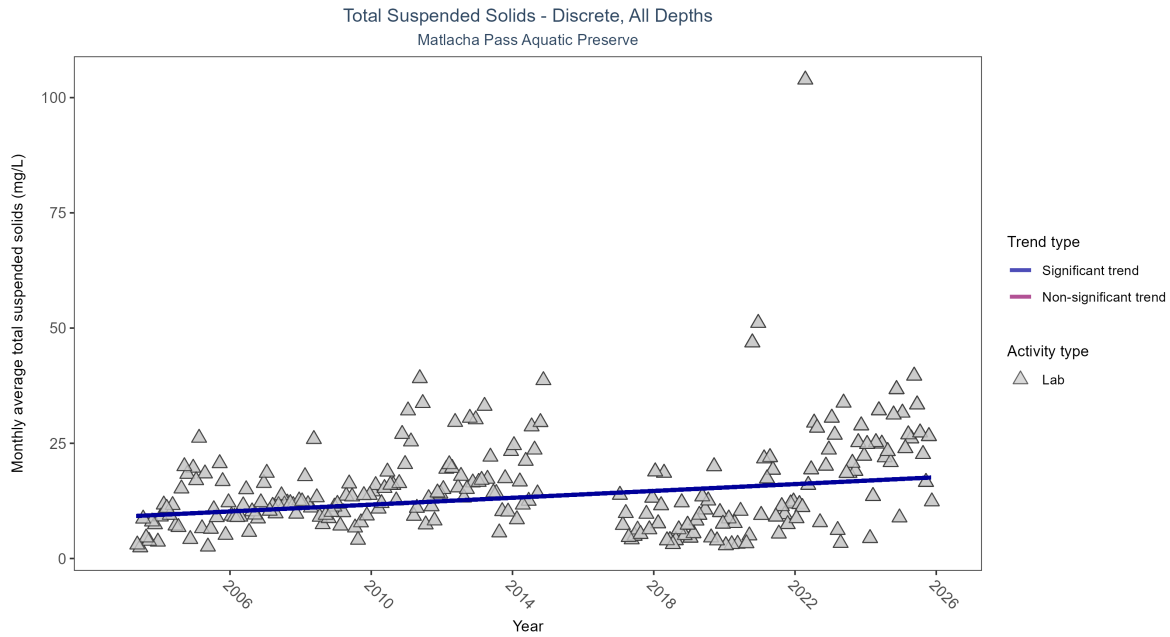


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	1184	21	2003 - 2025	11	0.19938	9.09711	0.37083	0

Monthly average total suspended solids increased by 0.37 mg/L per year, indicating a decrease in water clarity.

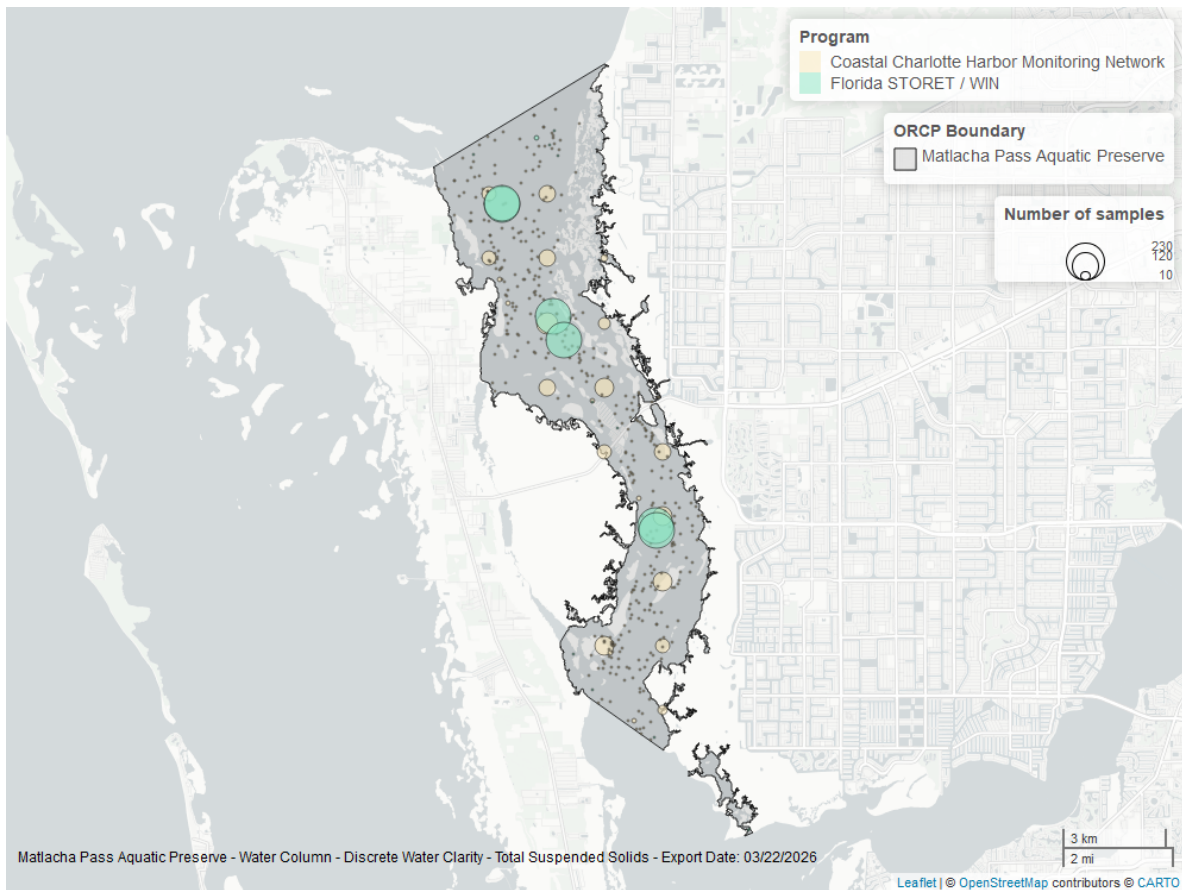


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

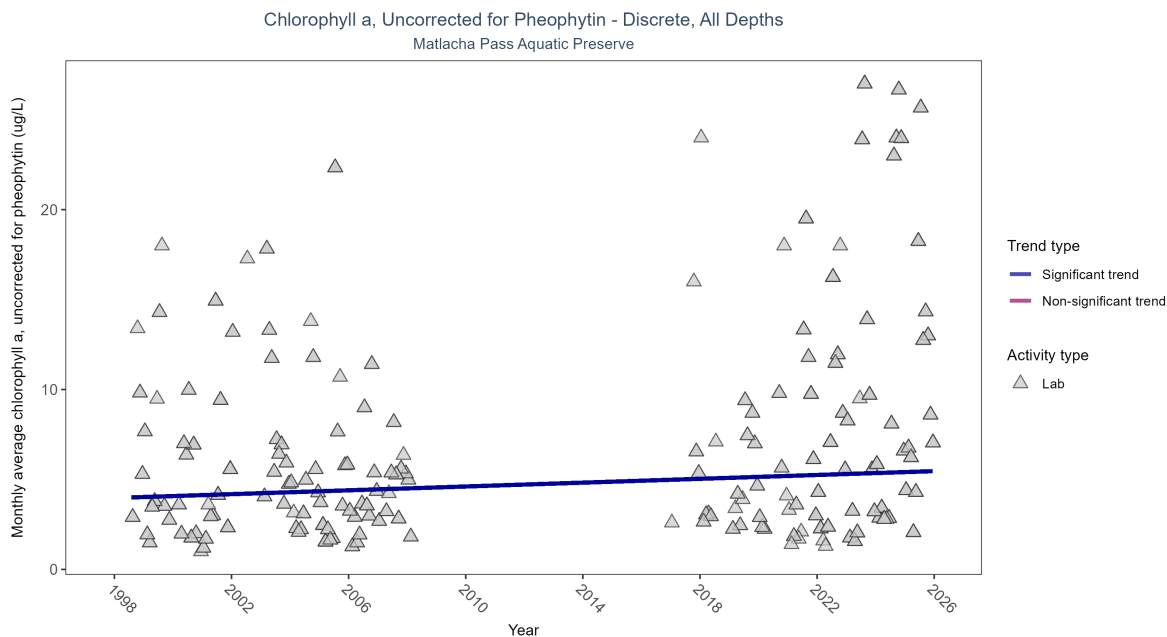


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	433	20	1998 - 2025	4.77	0.16534	3.97034	0.05333	0.0037

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.05 µg/L per year, indicating a decrease in water clarity.

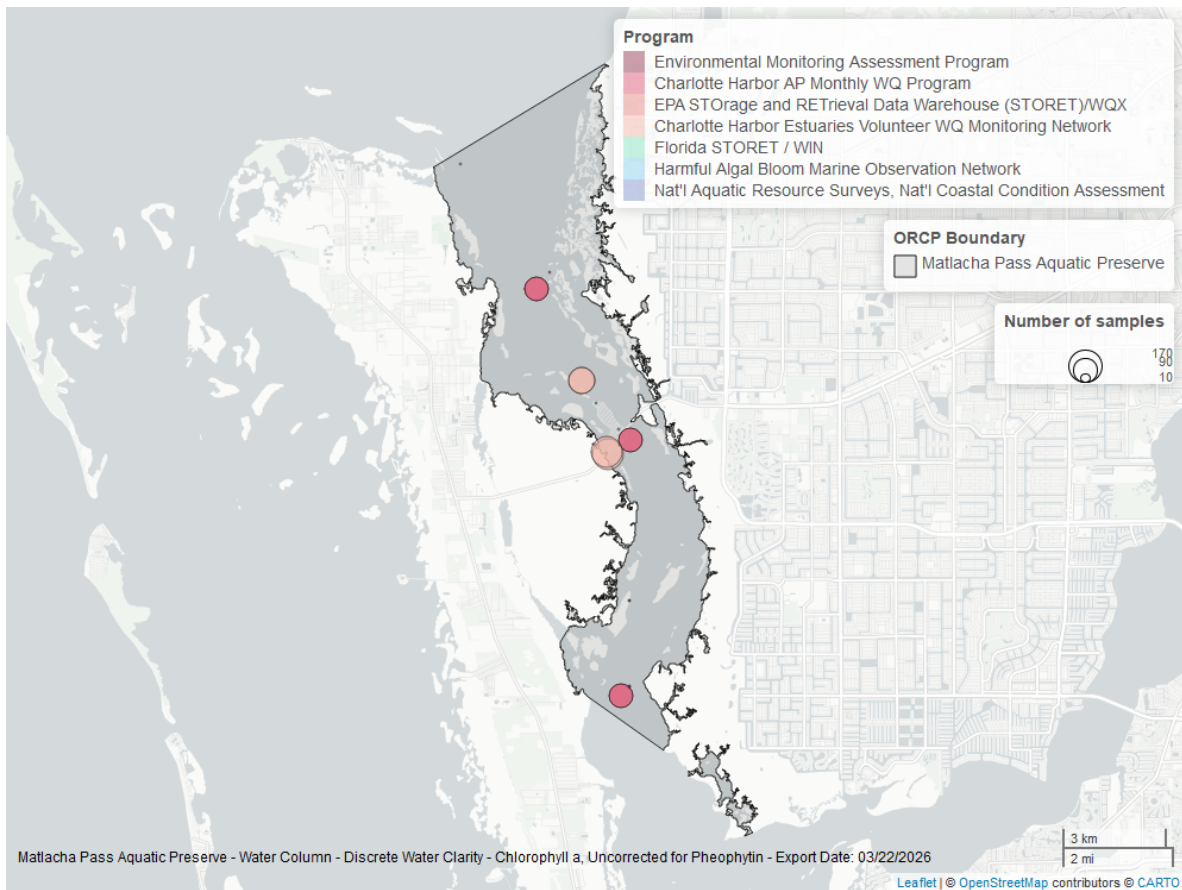


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

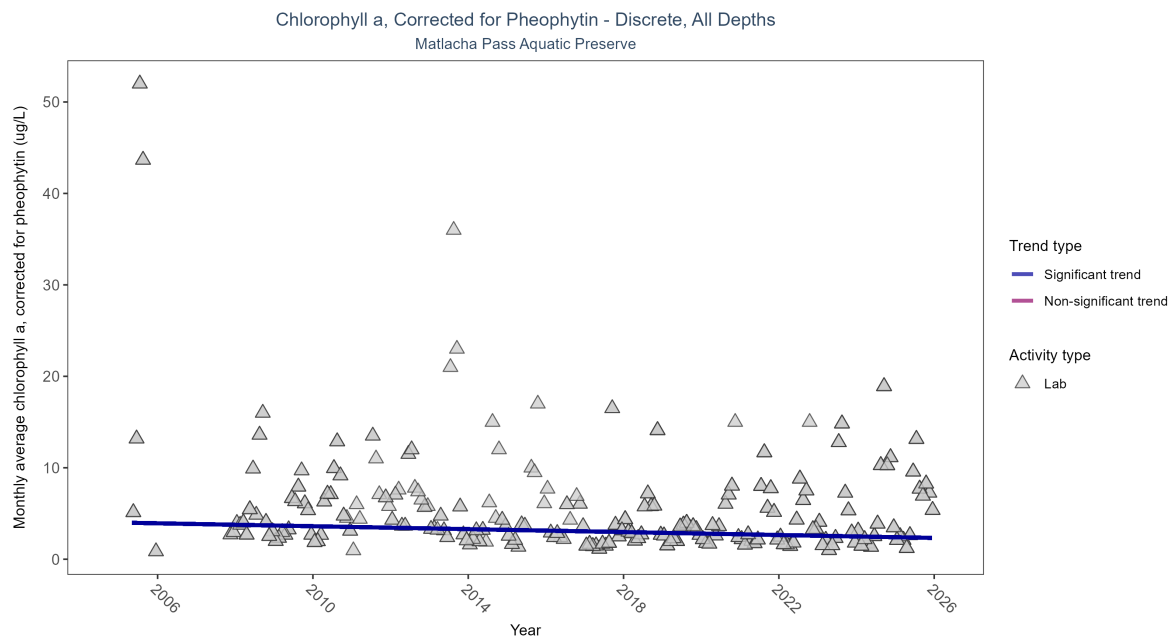


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	964	20	2005 - 2025	3	-0.19419	4.00722	-0.08	0.0001

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.08 $\mu\text{g/L}$ per year, indicating an increase in water clarity.

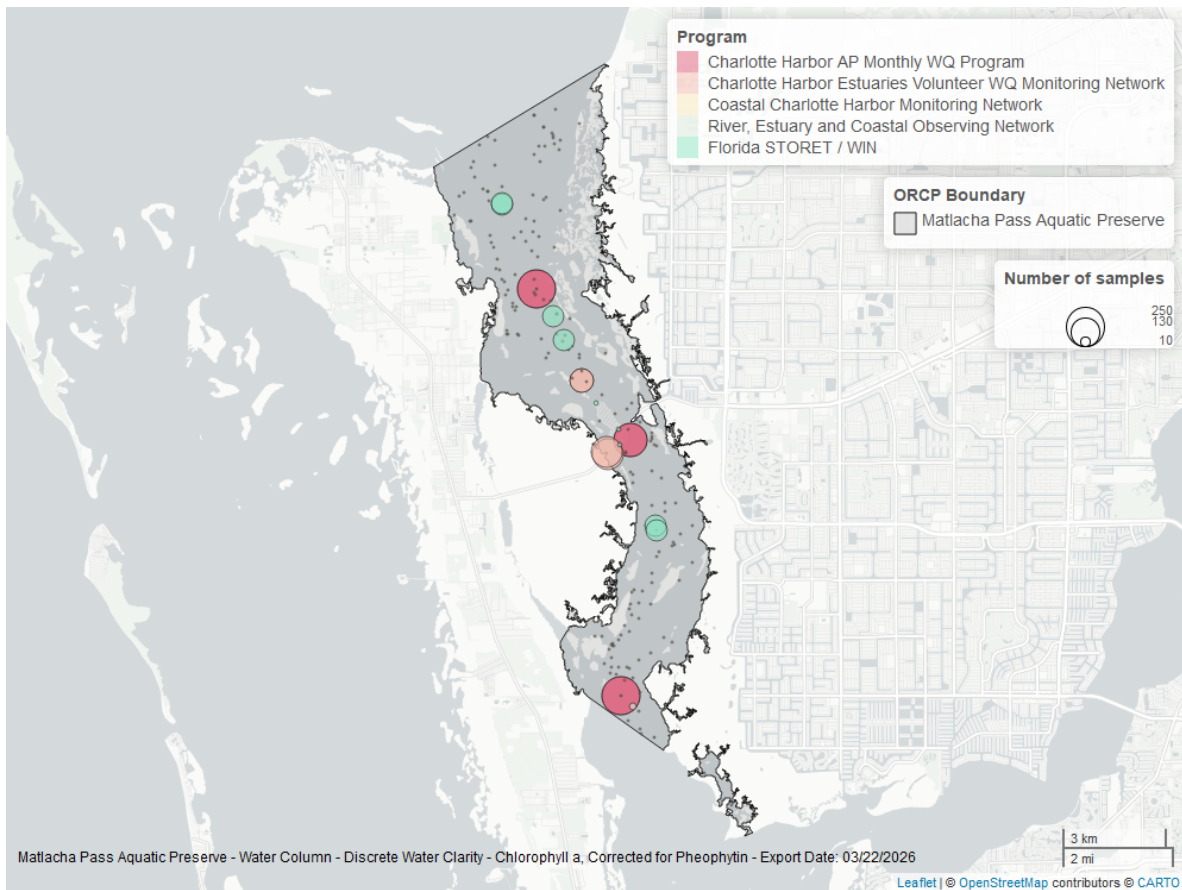


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

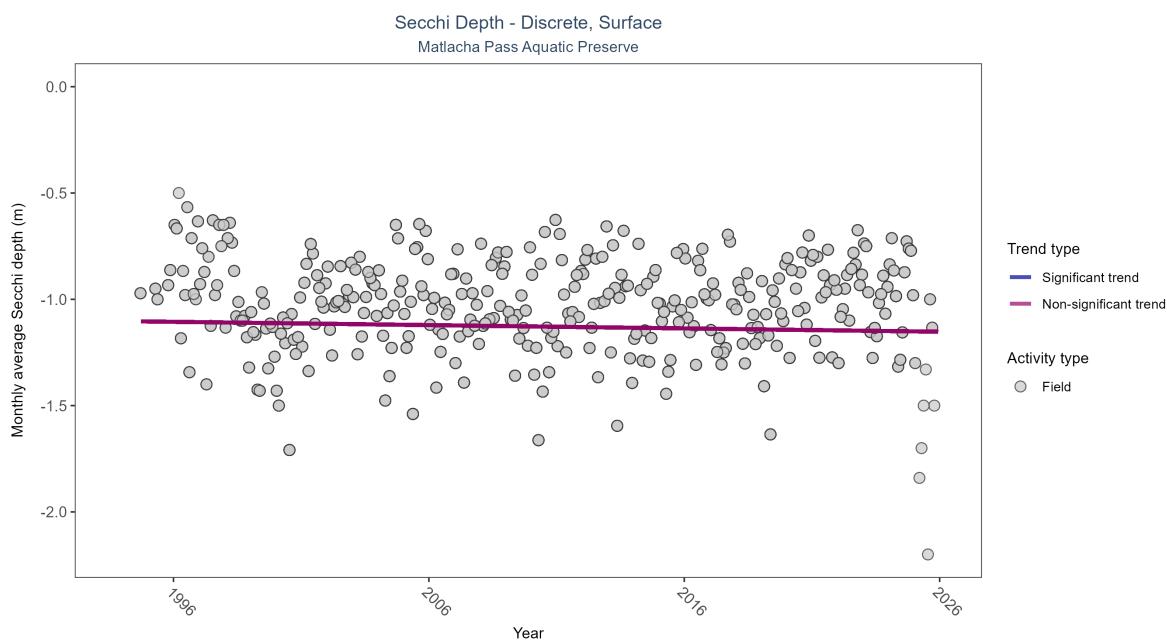


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	6666	32	1994 - 2025	-0.9	-0.03815	-1.10276	-0.00155	0.3062

Secchi depth showed no detectable trend between 1994 and 2025.

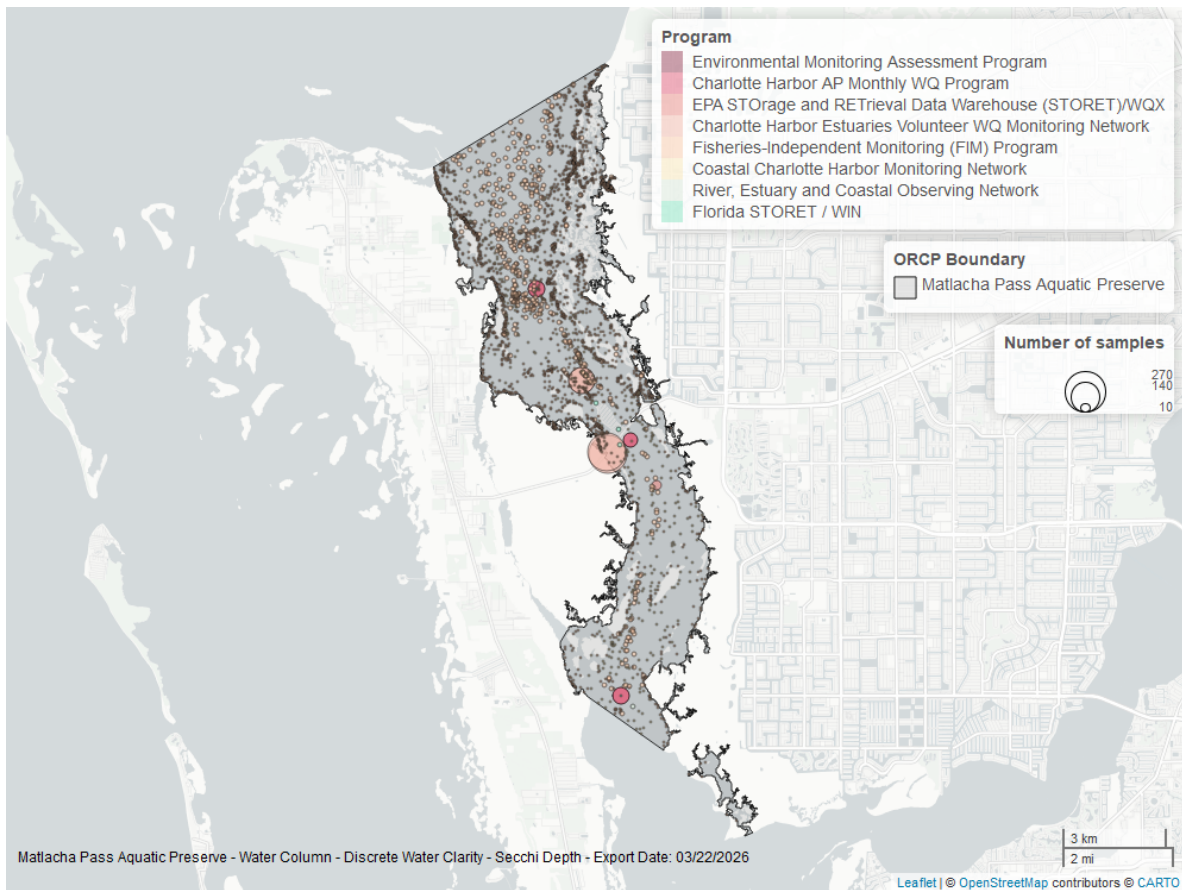


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

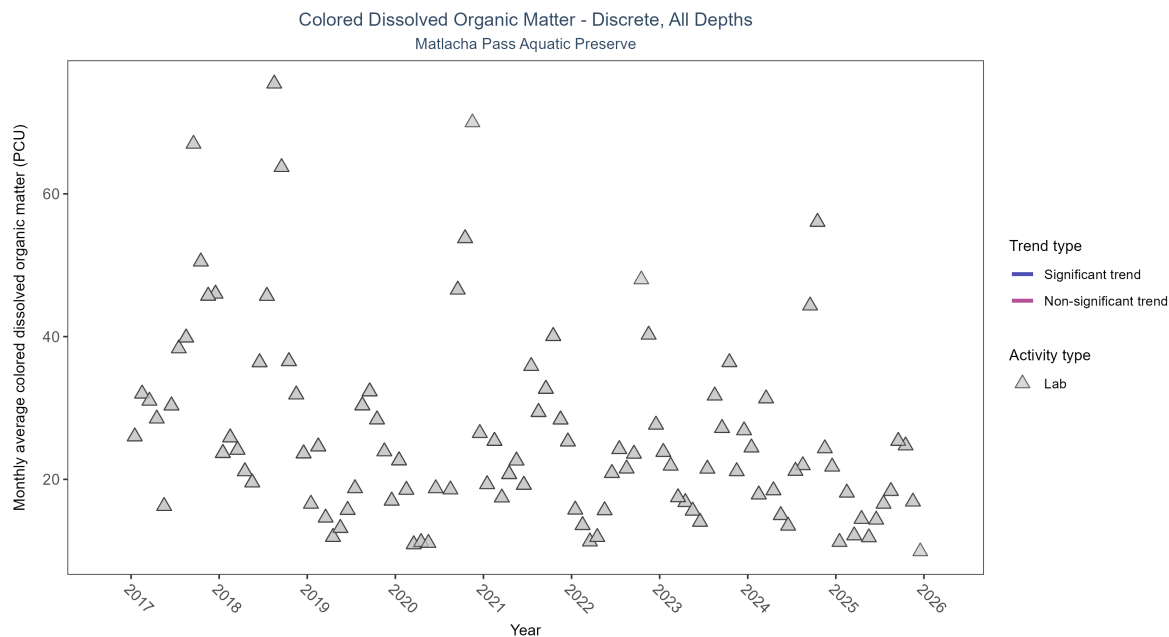


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	547	9	2017 - 2025	24.4	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

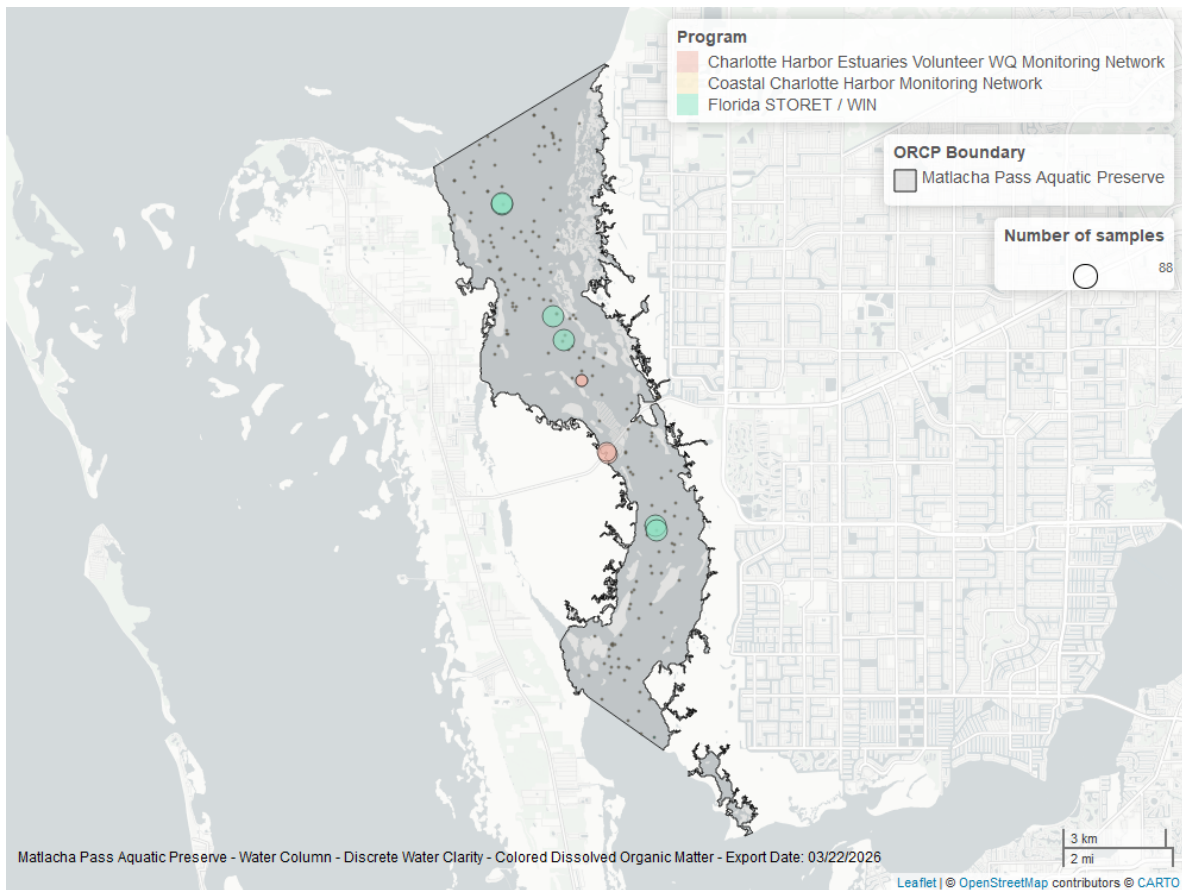


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Matlacha Pass Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.